Starfighter ANPC Guideline Document

Document #2 — ButterflyFX™ ANPC Design Suite

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Version: 1.0

Date: April 2026

Status: Authoritative Reference

"Store Geometry Not Bytes"

Classification: Title-Specific — Inherits from Document #1 (General ANPC Guideline Document)

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Manifold Surfaces $z=xy$ and $z=xy^2$ — Foundational Data-as-Geometry System

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Table of Contents

1. Introduction

1.1 Purpose and Scope

1.2 Inheritance Model

1.3 Starfighter Game Overview

2. Starfighter ANPC Role Roster

2.1 Role Taxonomy Table

2.2 Role Behavioral Profiles

2.3 Role Hierarchy and Chain of Command

3. Starfighter Identity Schema Extensions

3.1 Extended Identity Fields

3.2 Ship-Class Behavioral Modifiers

3.3 Weapon System Taxonomy

4. Combat Behavioral Rules

4.1 Engagement Rules of Engagement (ROE)

4.2 Combat State Machine

4.3 Formation Directives

4.4 Target Priority System

4.5 Morale and Attrition System

5. Starfighter Manifold Applications

5.1 Combat Manifold Mapping

5.2 Manifold-Driven Combat Decisions

5.3 Weapon Selection via Manifold

5.4 Manifold Drift in Extended Combat

6. Starfighter Communication Protocol

6.1 Radio Communication Model

6.2 Combat Callout Taxonomy

6.3 Communication Frequency Rules

7. Starfighter Phrase Pools

7.1 Title-Pool: Starfighter

7.2 Character-Pool Examples

7.3 Enemy Phrase Pools

8. Starfighter Composite String Assembly

8.1 Combat String Templates

8.2 Worked Assembly Examples

9. Starfighter Interaction Rules

9.1 Player-to-ANPC Combat Interactions

9.2 ANPC-to-ANPC Combat Interactions

9.3 Between-Mission Interactions

9.4 Disposition Shifts from Combat Events

10. Starfighter Scenario Vectors

10.1 Mission Type Scenario Templates

10.2 Dynamic Scenario Vector Modifiers

10.3 Scenario Vector Boundaries

11. Developer Implementation Notes — Starfighter Specific

11.1 ANPC Spawning and Initialization

11.2 Performance Budget

11.3 Combat AI Tuning Parameters

11.4 Save/Load Considerations

11.5 Multiplayer Considerations

12. Glossary — Starfighter Extensions

13. Document Index — Starfighter

14. References

15. Relational Diagrams

15.1 Chain of Command Flow

15.2 Combat State Machine Diagram

15.3 Weapon Selection Decision Tree

15.4 Scenario Vector Pipeline

15.5 ANPC Lifecycle Diagram

Appendix A: Complete Sample ANPC — Wingman "Hotshot" (Marcus Chen)

Appendix B: Complete Sample ANPC — Enemy Ace "Nightshade" (Unknown)

Appendix C: Complete Sample ANPC — Command Base Operator "Lighthouse" (Dr. Amara Okafor)

Appendix D: Mission Script Integration Example

Appendix E: Difficulty Scaling Matrix

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1. Introduction

1.1 Purpose and Scope

This document — the **Starfighter ANPC Guideline Document (Document #2)** — is the title-specific rulebook governing all Autonomous Non-Player Characters (ANPCs) within the *Starfighter* game title exclusively. It is the second document in the ButterflyFX™ ANPC Design Suite and operates under the authority of Kenneth Bingham.

Document #2 extends Document #1 (General ANPC Guideline Document) with combat-theater specifics, space-operations parameters, military-hierarchy definitions, weapon system taxonomies, and formation behavioral rules that are unique to the Starfighter title. All ANPCs in Starfighter must comply with both Document #1 and this document simultaneously.

> **IMPORTANT — Inheritance Rule**
>
> No rule in this document may contradict Document #1. Where this document is silent on a topic, Document #1 governs without exception. Where this document extends a Document #1 rule, the extension applies only within the Starfighter title context. Conflicts must be resolved in favor of Document #1 and escalated to the ButterflyFX™ Design Authority for reconciliation.

This document is intended for Starfighter developers, game designers, AI engineers, and QA engineers. All personnel working on ANPC behavior, dialogue, combat AI, or mission scripting within the Starfighter title must read and comply with this document in its entirety.

All ANPC systems in Starfighter operate under the ButterflyFX™ foundational philosophy: **"Store Geometry Not Bytes."** Behavioral data is encoded as positions on the manifold surfaces $z=xy$ and $z=xy^2$, not as discrete byte-level flags. This ensures continuous, smooth behavioral transitions and eliminates the brittle state boundaries that plague traditional NPC systems.

1.2 Inheritance Model

The ButterflyFX™ ANPC Design Suite follows a strict inheritance chain:

1. **Document #1 (Universal)** — General ANPC Guideline Document. Defines universal ANPC architecture, identity schema, personality vectors, disposition system, manifold mathematics, phrase pool structure, composite string assembly, and simulation-awareness framework. Applies to all ButterflyFX™ titles.
2. **Document #2 (Title-Specific: Starfighter)** — This document. Extends Document #1 with combat roles, weapon systems, formation behaviors, military hierarchy, space-operations communication protocols, and Starfighter-specific manifold applications. Applies only to the Starfighter title.

The following table identifies which Document #1 sections are inherited without modification and which are extended in this document:

Document #1 Section	Inherited As-Is	Extended in Doc #2	Doc #2 Reference
§1 — Introduction and Philosophy	Yes	No	—
§2 — ANPC Definition and Architecture	Yes	No	—
§3 — Identity Schema	Yes	Yes — extended fields	Section 3.1
§4 — Personality Vector System	Yes	Yes — role-specific ranges	Section 2.2
§5 — Disposition and Relationship System	Yes	Yes — combat event deltas	Section 9.4
§6 — Manifold Surface System ($z=xy$, $z=xy^2$)	Yes	Yes — combat mapping	Section 5
§7 — Scenario Vector Framework	Yes	Yes — mission templates	Section 10
§8 — Phrase Pool Architecture	Yes	Yes — title-pool added	Section 7

Document #1 Section	Inherited As-Is	Extended in Doc #2	Doc #2 Reference
§9 — Composite String Assembly	Yes	Yes — combat templates	Section 8
§10 — Simulation-Awareness Levels	Yes	No	—
§11 — Interaction Rules	Yes	Yes — combat interactions	Section 9
§12 — Developer Implementation Notes	Yes	Yes — Starfighter specifics	Section 11
§13 — Glossary	Yes	Yes — Starfighter terms	Section 12

1.3 Starfighter Game Overview

Starfighter is a space combat game in which the player pilots a starfighter through a campaign of missions involving dogfights, fleet engagements, escort duties, reconnaissance operations, ambush scenarios, and strategic offensive operations. The game combines real-time flight combat with narrative progression, inter-mission planning, and persistent ANPC relationships.

ANPCs in Starfighter serve the following functional roles:

- **Wingmen** — allied pilots who fly alongside the player, follow formation orders, and engage targets on command or autonomously based on ROE
- **Squadron Leaders** — command-level ANPCs who issue tactical orders and coordinate multi-ship maneuvers
- **Command Base Operators** — remote support personnel providing mission briefings, intelligence relay, and extraction coordination
- **Enemy Combatants** — hostile pilots ranging from standard pilots to elite aces with unique personalities and combat signatures
- **Fleet Commanders** — strategic-level officers issuing campaign directives with rare direct player interaction

- **Civilian Contacts** — mission givers, informants, traders, and non-combatants encountered between missions
- **Mercenaries** — allegiance-fluid characters motivated by payment and self-interest who may switch sides mid-campaign
- **Drone Controllers** — specialists commanding autonomous drone swarms for tactical support
- **Ship Mechanics / Engineers** — repair, upgrade, and maintenance personnel encountered between missions

Starfighter is part of the ButterflyFX™ proof-of-concept lineup, with planned commercial release on Steam, Google Play, and major game platforms. All ANPC behavioral data in Starfighter is encoded using the manifold surfaces $z=xy$ and $z=xy^2$, embodying the "Store Geometry Not Bytes" philosophy. Behavioral states are not stored as discrete flags or bitmasks; they are stored as continuous coordinates on these manifold surfaces, enabling smooth interpolation and emergent behavioral nuance.

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2. Starfighter ANPC Role Roster

2.1 Role Taxonomy Table

The following table defines all ANPC roles within the Starfighter title. Each role is assigned a unique Role ID, a category alignment (Allied, Enemy, or Neutral), and an expected simulation-awareness level range per Document #1 §10.

Role ID	Role Name	Category	Description	Typical Rank Range	Sim-Aware Level	Key Behavioral Traits
SF-WING	Wingman	Allied	Flies alongside player, follows formation	Ensign – Lieutenant	0 – 2	Loyal, responsive, formation-

Role ID	Role Name	Category	Description	Typical Rank Range	Sim-Aware Level	Key Behavioral Traits
			orders, engages targets on command or autonomously per ROE			disciplined, variable aggression
SF-SQLDR	Squadron Leader	Allied	Commands a flight group, issues tactical orders, coordinates multi-ship maneuvers	Captain – Commander	0 – 2	Authoritative, tactical, decisive, high communication frequency
SF-CMDOP	Command Base Operator	Allied	Remote support: mission briefings, intel relay, extraction coordination, strategic updates	Lieutenant – Colonel	0 – 1	Calm, analytical, high-information density, non-combatant
SF-EACE	Enemy Ace	Enemy	Elite enemy pilot with unique personality, taunts, signature combat style, and narrative significance	Elite – Commander	0 – 3	Aggressive, taunting, distinctive maneuvers, high skill ceiling

Role ID	Role Name	Category	Description	Typical Rank Range	Sim-Aware Level	Key Behavioral Traits
SF-EPLT	Enemy Pilot	Enemy	Standard enemy combatant with varying skill levels, generic personality, and expendable narrative role	Recruit – Veteran	0 – 1	Variable aggression, role-dependent tactics, standardized comms
SF-FADM	Fleet Admiral	Allied	Strategic-level commander issuing campaign directives; rare direct interaction, high narrative authority	Admiral	1 – 3	Commanding, strategic, measured, rare communication
SF-CIV	Civilian Contact	Neutral	Mission givers, informants, traders, non-combatants encountered in stations, ports, and comm channels	N/A (civilian)	0 – 2	Variable personality, non-aggressive, information-oriented
SF-MERC	Mercenary	Neutral	Allegiance-fluid pilot	Freelancer – Elite	0 – 2	Self-interested,

Role ID	Role Name	Category	Description	Typical Rank Range	Sim-Aware Level	Key Behavioral Traits
			motivated by payment and self-interest; may switch sides based on loyalty_score			pragmatic, variable loyalty, opportunistic
SF-DRONE	Drone Controller	Allied	Commands autonomous drone swarms for tactical support; operates from carrier or capital ship	Specialist – Lieutenant	0 – 1	Systematic, low emotion, high tactical precision, low charisma
SF-MECH	Ship Mechanic / Engineer	Allied	Repair, upgrade, and maintenance interactions between missions; provides ship readiness status	Technician – Chief Engineer	0 – 1	Practical, detail-oriented, straightforward, non-combatant

2.2 Role Behavioral Profiles

Each role has a default behavioral profile specifying the expected ranges for Document #1's five personality axes (Openness, Conscientiousness, Extraversion,

Agreeableness, Neuroticism), plus disposition, charisma, motivators, and communication patterns. All values are floats on a [0.0–1.0] scale unless otherwise noted.

SF-WING — Wingman

Attribute	Range / Default
Default Disposition	0.55 – 0.80 (friendly, trusting)
Openness	0.3 – 0.6
Conscientiousness	0.5 – 0.8
Extraversion	0.3 – 0.7
Agreeableness	0.5 – 0.9
Neuroticism	0.1 – 0.5
Charisma	0.3 – 0.6
Typical Motivators	Duty, camaraderie, survival, player approval
Communication Frequency	Medium (every 8–15 seconds during combat)
Communication Style	Responsive, brevity-code-heavy, acknowledges orders
Combat Engagement	Follows player lead; engages player-designated targets first; autonomous engagement on Weapons Free

SF-SQLDR — Squadron Leader

Attribute	Range / Default
Default Disposition	0.50 – 0.70 (professional, measured)
Openness	0.4 – 0.7
Conscientiousness	0.7 – 0.95
Extraversion	0.5 – 0.8
Agreeableness	0.4 – 0.7
Neuroticism	0.1 – 0.3

Attribute	Range / Default
Charisma	0.5 – 0.8
Typical Motivators	Mission success, unit survival, command authority, tactical excellence
Communication Frequency	High (every 5–10 seconds during combat)
Communication Style	Directive, tactical, uses formation and flanking commands
Combat Engagement	Coordinates squadron; engages high-value targets; issues break and reform orders

SF-CMDOP — Command Base Operator

Attribute	Range / Default
Default Disposition	0.60 – 0.80 (supportive, stable)
Openness	0.5 – 0.8
Conscientiousness	0.8 – 0.95
Extraversion	0.3 – 0.5
Agreeableness	0.6 – 0.8
Neuroticism	0.05 – 0.2
Charisma	0.4 – 0.7
Typical Motivators	Mission success, pilot safety, information accuracy
Communication Frequency	Event-driven (briefings, intel updates, extraction calls)
Communication Style	Calm, data-rich, uses precise coordinates and ETA values
Combat Engagement	Non-combatant; provides tactical overlays and threat assessments

SF-EACE — Enemy Ace

Attribute	Range / Default
Default Disposition	-0.80 – -0.40 (hostile, contemptuous)
Openness	0.3 – 0.7
Conscientiousness	0.5 – 0.9
Extraversion	0.5 – 0.9
Agreeableness	0.1 – 0.3
Neuroticism	0.1 – 0.6
Charisma	0.6 – 0.9
Typical Motivators	Glory, rivalry, proving superiority, personal vendetta
Communication Frequency	Medium-high (taunts, grudging respect during combat)
Communication Style	Personal, taunting, dramatic, references player by callsign
Combat Engagement	Prioritizes player; uses signature maneuvers; will not retreat until damage_threshold exceeded

SF-EPLT — Enemy Pilot

Attribute	Range / Default
Default Disposition	-0.70 – -0.30 (hostile, professional)
Openness	0.2 – 0.5
Conscientiousness	0.3 – 0.7
Extraversion	0.2 – 0.5
Agreeableness	0.2 – 0.4
Neuroticism	0.2 – 0.7
Charisma	0.1 – 0.4
Typical Motivators	Orders, survival, unit cohesion
Communication Frequency	Low (standardized callouts only)
Communication Style	Brevity-code-only, translated/intercepted, generic
Combat Engagement	Follows ROE from enemy command;

Attribute	Range / Default
	engages nearest threat; retreats when morale is low

SF-FADM — Fleet Admiral

Attribute	Range / Default
Default Disposition	0.40 – 0.65 (professional, detached)
Openness	0.4 – 0.6
Conscientiousness	0.85 – 0.98
Extraversion	0.4 – 0.6
Agreeableness	0.3 – 0.5
Neuroticism	0.05 – 0.15
Charisma	0.7 – 0.95
Typical Motivators	Campaign victory, fleet preservation, strategic objectives
Communication Frequency	Very low (mission briefings, campaign-critical moments only)
Communication Style	Commanding, deliberate, uses strategic terminology, addresses groups
Combat Engagement	Non-combatant (command level); issues fleet-wide directives

SF-CIV — Civilian Contact

Attribute	Range / Default
Default Disposition	0.30 – 0.70 (variable, context-dependent)
Openness	0.4 – 0.9
Conscientiousness	0.2 – 0.6
Extraversion	0.3 – 0.8
Agreeableness	0.4 – 0.8

Attribute	Range / Default
Neuroticism	0.2 – 0.6
Charisma	0.2 – 0.7
Typical Motivators	Self-interest, information trade, survival, profit
Communication Frequency	Dialogue-driven (mission briefings, trade interactions)
Communication Style	Conversational, informal, variable vocabulary
Combat Engagement	Non-combatant; flees or panics under fire; may provide distress calls

SF-MERC — Mercenary

Attribute	Range / Default
Default Disposition	0.10 – 0.50 (guarded, transactional)
Openness	0.3 – 0.6
Conscientiousness	0.3 – 0.6
Extraversion	0.4 – 0.7
Agreeableness	0.2 – 0.5
Neuroticism	0.2 – 0.5
Charisma	0.4 – 0.7
Typical Motivators	Payment, reputation, survival, self-preservation above all
Communication Frequency	Medium (negotiation-heavy, combat-callouts when engaged)
Communication Style	Casual, mercenary slang, references payment and contracts
Combat Engagement	Engages per contract terms; disengages when loyalty_score drops below 0.3; may switch sides

SF-DRONE — Drone Controller

Attribute	Range / Default
Default Disposition	0.50 – 0.65 (neutral, task-focused)
Openness	0.2 – 0.4
Conscientiousness	0.8 – 0.95
Extraversion	0.1 – 0.3
Agreeableness	0.4 – 0.6
Neuroticism	0.1 – 0.3
Charisma	0.1 – 0.3
Typical Motivators	Tactical efficiency, drone swarm integrity, system optimization
Communication Frequency	Low (status reports, deployment confirmations)
Communication Style	Technical, clipped, system-status oriented, minimal personality
Combat Engagement	Deploys and manages drone swarms; does not personally engage; retreats if command vessel threatened

SF-MECH — Ship Mechanic / Engineer

Attribute	Range / Default
Default Disposition	0.55 – 0.75 (amiable, matter-of-fact)
Openness	0.3 – 0.5
Conscientiousness	0.7 – 0.9
Extraversion	0.3 – 0.6
Agreeableness	0.5 – 0.7
Neuroticism	0.1 – 0.3
Charisma	0.2 – 0.5
Typical Motivators	Ship integrity, craftsmanship, pilot safety, professional pride
Communication Frequency	Between-mission dialogue; repair status updates
Communication Style	Technical jargon, gruff or warm

Attribute	Range / Default
	depending on personality, references ship parts
Combat Engagement	Non-combatant; hangar/station interactions only

2.3 Role Hierarchy and Chain of Command

The Starfighter military hierarchy governs order propagation, authority, and insubordination thresholds. Orders flow downward through the chain; reports and requests flow upward.

Allied Force Hierarchy

```
Fleet Admiral (SF-FADM)
│
├──► Squadron Leader (SF-SQLDR)
│   │
│   ├──► Wingman (SF-WING)
│   │
│   └──► Drone Controller (SF-DRONE)
│
└──► Command Base Operator (SF-CMDOP)
│
└──► [advisory role to all combat units]

Support Staff (non-chain):
Ship Mechanic / Engineer (SF-MECH) — reports to base command
```

Enemy Force Hierarchy

```
Enemy Fleet Command (off-screen authority)
│
├──► Enemy Ace (SF-EACE)
│   │
│   └──► Enemy Pilot (SF-EPLT) [subordinate wing]
│
└──► Enemy Pilot (SF-EPLT) [standard groups]
```

Neutral Entities (Outside Chain)

- **Civilian Contact (SF-CIV)** — outside all military chains; interacts through dialogue only

- **Mercenary (SF-MERC)** — outside formal chain; bound by contract terms; accepts player orders when loyalty_score > 0.5

Order Propagation Rules

3. **Downward propagation:** Orders from Fleet Admiral propagate to Squadron Leaders, then to Wingmen and Drone Controllers. Each level may interpret the order within their tactical context per Document #1 §11.3.
4. **Player authority:** The player holds functional authority equivalent to Squadron Leader when assigned squadron command, or Wingman when assigned as a flight member. Player ROE overrides propagate only to directly assigned wingmen.
5. **Order refusal threshold:** An ANPC may refuse an order when the order's danger level exceeds (1.0 - neuroticism) × disposition. If refusal triggers, the ANPC selects a refusal phrase from the phrase pool and provides an alternative action.
6. **Insubordination mechanics:** Repeated order refusals (3+ within a single mission) flag the ANPC for "insubordination state." This triggers a post-mission debrief event and a disposition penalty of -0.1 against the ordering authority.

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3. Starfighter Identity Schema Extensions

3.1 Extended Identity Fields

The following fields extend Document #1's Identity Schema (§3) with Starfighter-specific data. These fields are required for all Starfighter ANPCs in addition to the base fields defined in Document #1. All fields are stored as coordinates or parameters within the manifold system — "Store Geometry Not Bytes."

Field Name	Type	Range / Format	Description	Persistence
ship_class	string (enum)	Interceptor, Bomber, Heavy Fighter, Scout, Capital Ship	Assigned ship type; determines behavioral modifier profile (§3.2)	Mission-set
squadron_id	string	Alphanumeric ID (e.g., "SQ-07")	Squadron assignment identifier; links ANPC to squadron unit	Campaign-persistent
squadron_name	string	Display string (e.g., "Shadow Wolves")	Human-readable squadron name for dialogue and UI	Campaign-persistent
callsign	string	1–2 word tactical name (e.g., "Frostbite")	Tactical radio callsign; inherited from Doc #1 but **mandatory** in Starfighter	Permanent
combat_rating	float	[0.0 – 1.0]	Overall combat effectiveness score; affects accuracy, reaction time, and tactical sophistication	Campaign-persistent
weapon_loadout	array of strings	List from weapon taxonomy (§3.3)	Currently equipped weapons; affects phrase triggers and combat behavior	Mission-set
flight_hours	integer	[0 – 99999]	Simulated flight experience in hours; affects skill curves and dialogue	Campaign-persistent

Field Name	Type	Range / Format	Description	Persistence
			references	
kill_count	integer	[0 – 99999]	Tracked confirmed eliminations; affects reputation, disposition modifiers, and dialogue tone	Campaign-persistent
loyalty_score	float	[0.0 – 1.0]	For mercenaries and allegiance-fluid roles; below 0.3 triggers side-switch evaluation	Campaign-persistent
damage_threshold	float	[0.0 – 1.0]	Hull damage level (as fraction of max HP) triggering behavioral shifts: retreat, desperation, or surrender	Design-set
preferred_tactic	enum	aggressive, defensive, evasive, balanced, ambush, support	Default tactical approach; modifies manifold input weighting	Design-set
formation_position	enum	lead, wing_left, wing_right, rear_guard, free_roam	Assigned position in formation flying; affects spatial behavior and target selection	Mission-dynamic

3.2 Ship-Class Behavioral Modifiers

Ship class applies a behavioral modifier profile that adjusts ANPC behavior at the manifold level. These modifiers are multiplied against the ANPC's base personality vector before manifold lookup, ensuring that a cautious pilot in a Bomber behaves differently from a cautious pilot in an Interceptor. This is geometry-driven modification — the ship class warps the personality vector's position on the manifold surface.

Ship Class	Aggression Ceiling	Speed Preference	Damage Tolerance	Maneuverability	Engagement Style	Comms Frequency Modifier
Interceptor	0.9 (high)	0.9 (very high)	0.3 (low)	0.9 (very high)	Fast attack, hit-and-run, pursuit	×1.2 (more frequent)
Bomber	0.5 (moderate)	0.3 (low)	0.6 (moderate)	0.3 (low)	Target-focused, bombing runs, methodical approach	×0.8 (less frequent)
Heavy Fighter	0.7 (moderate-high)	0.5 (moderate)	0.8 (high)	0.5 (moderate)	Sustained engagement, area denial, escort	×1.0 (standard)
Scout	0.3 (low)	0.8 (high)	0.2 (very low)	0.8 (high)	Evasive, recon-focused, low engagement threshold	×1.5 (high — intel relay)
Capital Ship	0.4 (low-moderate)	0.1 (very low)	0.95 (very high)	0.1 (very low)	Command-oriented, strategic, area-of-effect weaponry	×0.6 (infrequent, strategic only)

3.3 Weapon System Taxonomy

The following weapon systems define the armaments ANPCs reference, select, and interact with during combat. Each weapon has an associated ANPC usage context and phrase triggers that connect to the phrase pool system (Section 7). Weapon selection is driven by manifold output (Section 5.3).

Weapon ID	Name	Type	Range Profile	Damage Profile	Ammo Type	ANPC Usage Context	Phrase Triggers
WPN-LAS	Lasers	Energy (direct fire)	Short-Medium (200–800m)	Moderate; sustained DPS	Infinite (energy-based)	Primary engagement tool; default weapon; used in all combat states	"Guns guns guns", "Firing lasers", "Hosing him down"
WPN-SCT	Scatter Shot	Projectile (spread)	Close (50–300m)	High per-hit at close range; disperses with distance	Limited (120 rounds default)	Close-range defense, pursuing evasive targets, point-blank finishing	"Switching to scatter", "Close enough—scatter out", "Shredding"
WPN-PTN	Proton Torpedoes	Heavy ordnance (guided)	Medium-Long (500–2000m)	Very high; single-target devastation	Limited (4–8 torpedoes default)	Lock-on required (2.5s); used against capital ships, heavy targets, high-value	"Fox Three", "Torpedo away", "Tone—firing", "Good lock, good

Weapon ID	Name	Type	Range Profile	Damage Profile	Ammo Type	ANPC Usage Context	Phrase Triggers
						contacts	lock"
WPN-EMP	EMP Burst	Area-of-effect (electronic)	Radius (400m sphere)	No hull damage; disables shields and systems for 6–10 seconds	Limited (2 charges default)	Coordinated timing; warn allies before deployment; used to create vulnerability windows	"EMP hot — all friendlies clear the zone", "Popping EMP", "Systems kill in three"

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4. Combat Behavioral Rules

4.1 Engagement Rules of Engagement (ROE)

Rules of Engagement (ROE) define when ANPCs may engage, must engage, or must not engage hostile contacts. ROE is a command-level directive that propagates through the chain of command (Section 2.3). The Starfighter title defines three ROE levels:

ROE Level	Code	Definition	ANPC Behavior
Weapons Free	ROE-FREE	Engage any hostile contact at will without requiring authorization	ANPCs autonomously acquire and engage targets based on target priority system (§4.4). Maximum aggression

ROE Level	Code	Definition	ANPC Behavior
			allowed.
Weapons Tight	ROE-TIGHT	Engage only positively identified hostile targets or targets designated by command/player	ANPCs must confirm target is hostile (Bandit classification) before engaging. Self-defense is always permitted.
Weapons Hold	ROE-HOLD	Do not fire except in immediate self-defense	ANPCs maintain defensive posture. Will only fire if directly fired upon. Used during stealth and diplomatic missions.

ROE Propagation: ROE is set at the highest applicable authority level and propagates downward. Fleet Admiral sets campaign-level ROE; Squadron Leader may tighten (not loosen) for their squadron; player may issue ROE to assigned wingmen within the bounds set by their Squadron Leader.

Player ROE Overrides: The player may issue ROE changes to their direct wingmen at any time. However, if the player's ROE contradicts a higher-authority ROE (e.g., player issues Weapons Free while Fleet Admiral has ordered Weapons Hold), wingmen with high Conscientiousness (>0.7) will refuse the override and reference the standing order. Wingmen with low Conscientiousness (<0.4) may comply with a disposition penalty toward the Squadron Leader.

4.2 Combat State Machine

Every ANPC in Starfighter maintains a combat state that governs their behavioral priorities, communication patterns, and manifold input weighting. The following eight states are defined with their transition triggers:

State	ID	Description	Behavioral Priorities	Comm Pattern
Patrol	CS-PATROL	No threats detected;	Scanning, formation	Low frequency;

State	ID	Description	Behavioral Priorities	Comm Pattern
		routine flight operations	maintenance, idle banter, fuel/ammo monitoring	casual chatter; status checks
Alert	CS-ALERT	Threat detected but not yet identified or engaged	Shields raised, weapons armed, formation tightens, sensor focus	Medium frequency; contact reports; Bogey calls
Engaged	CS-ENGAGED	Active combat; firing weapons, tracking targets	Target tracking, weapon firing, maneuver execution, threat assessment	High frequency; combat callouts; kill confirmations
Evasive	CS-EVASIVE	Under heavy fire; executing evasive maneuvers	Survival-first, break maneuvers, requesting support, chaff/flare deployment	Very high frequency; "Break" calls; support requests
Damaged	CS-DAMAGED	Hull integrity below damage_threshold; systems degrading	Behavioral degradation, reduced accuracy, limited maneuvers, desperation attacks or retreat evaluation	High frequency; damage reports; system failure warnings
Retreating	CS-RETREAT	Disengaging from combat; heading to safe zone	Maximum speed toward extraction point, minimal engagement, may re-engage if pursued and cornered	Medium frequency; retreat announcements; ETA to safe zone
Disabled	CS-DISABLED	Systems offline;	No combat capability;	Low frequency;

State	ID	Description	Behavioral Priorities	Comm Pattern
		drifting; no propulsion or weapons	may surrender, broadcast distress, or trigger self-destruct if Neuroticism > 0.7	distress beacon; surrender offer; final words
Destroyed	CS-DESTROYED	Terminal state; ANPC eliminated	Final transmission (if any), debris field spawned, kill credit assigned	Single final transmission (optional based on personality); silence

State Transition Table

From State	To State	Trigger Condition
Patrol	Alert	Unidentified contact within sensor range (2000m)
Alert	Patrol	Contact classified as friendly or lost; no contacts for 15 seconds
Alert	Engaged	Contact classified as Bandit AND ROE permits engagement
Engaged	Evasive	Incoming fire exceeds evasion threshold (3+ hits in 2 seconds)
Engaged	Damaged	Hull integrity drops below damage_threshold
Engaged	Patrol	All hostile contacts destroyed or disengaged; no contacts for 20 seconds
Evasive	Engaged	Successfully evaded; attacker shaken or destroyed; morale > 0.4

From State	To State	Trigger Condition
Evasive	Damaged	Hull integrity drops below damage_threshold during evasion
Evasive	Retreating	Morale < 0.3 AND evasion unsuccessful for 5+ seconds
Damaged	Retreating	Morale < 0.3 OR hull integrity < 0.15
Damaged	Engaged	Morale > 0.5 AND hull integrity > damage_threshold × 0.5 (desperation fight)
Damaged	Disabled	Hull integrity < 0.05 OR critical system failure (engines, weapons both offline)
Retreating	Engaged	Pursued AND cornered (no escape vector) AND morale > 0.2
Retreating	Disabled	Hull integrity reaches 0.05 during retreat
Disabled	Destroyed	Additional damage received OR self-destruct timer expires
Any State	Destroyed	Hull integrity reaches 0.0

4.3 Formation Directives

Formation flying is a core behavioral system in Starfighter. Formations are assigned by Squadron Leaders or the player and determine the spatial arrangement, target responsibility zones, and communication patterns of a flight group.

Standard Formations

Formation	ID	Description	Best Use	Max Ships
V-Formation	FRM-VEE	Leader at apex; wingmen	Standard patrol, balanced	5

Formation	ID	Description	Best Use	Max Ships
		spread behind at 45° angles	offense/defense	
Diamond	FRM-DIA	Lead, left, right, and rear guard positions; 360° coverage	Escort missions, high-threat environments	4
Line Abreast	FRM-LINE	All ships in a horizontal line; maximum forward firepower	Assault runs, strafing, overwhelming force	6
Echelon Left	FRM-ECL	Diagonal stagger to the left of the leader	Flanking approaches, oblique attack angles	4
Echelon Right	FRM-ECR	Diagonal stagger to the right of the leader	Flanking approaches, oblique attack angles	4
Spread	FRM-SPR	Wide dispersion; maximum sensor coverage; minimal mutual support	Reconnaissance, search patterns, ambiguity-reduction	8
Trail	FRM-TRL	Single-file behind leader; minimum cross-section	Stealth approach, narrow passages, asteroid fields	6

Formation Rules

- **Assignment:** Formation is assigned based on role and ship class. Interceptors prefer wing positions; Bombers prefer rear guard; Scouts prefer free_roam when in Spread formation.

- **Break conditions:** Formation automatically breaks when any member transitions to CS-ENGAGED, CS-EVASIVE, or CS-DAMAGED. The break is announced via combat callout ("Breaking formation!").
- **Reformation:** After all hostiles are neutralized and all members return to CS-PATROL, the Squadron Leader issues a "Reform on me" directive. Ships return to assigned positions over 8–12 seconds.
- **Formation tightness:** Controlled by the designer-exposed "formation_tightness" parameter [0.0–1.0], where 0.0 = maximum dispersion within formation type and 1.0 = parade-tight spacing. Default: 0.5.

4.4 Target Priority System

ANPCs select targets using a weighted priority score calculated from the following factors. The target with the highest priority score is selected as the primary target. This score is recalculated every 0.5 seconds during combat states.

Factor	Weight	Description
Player Designation	0.40	If the player has designated this target, priority increases dramatically
Threat Level	0.25	Based on target's combat_rating × current aggression toward this ANPC
Proximity	0.15	Inverse distance; closer targets score higher
Opportunity	0.10	Target vulnerability: low shields, damaged hull, exposed flank
Role-Specific Bias	0.10	See role-specific overrides below

Role-Specific Target Priority Overrides

Role	Priority Override
SF-WING (Wingman)	Prioritize player's designated target (+0.3 bias); if no designation, engage nearest threat to player
SF-SQLDR (Squadron Leader)	Prioritize highest-threat target to squadron; distribute targets among wingmen before self-engaging
SF-EACE (Enemy Ace)	Prioritize player above all other targets (+0.5 bias to player); ignore low-value targets unless no player contact
SF-EPLT (Enemy Pilot)	Engage nearest target; no strong preference bias; follows enemy squadron leader's designation if present
SF-MERC (Mercenary)	Prioritize targets that threaten the mercenary personally (+0.2 self-preservation bias)
SF-DRONE (Drone Controller)	Distributes drone swarm across multiple targets; prioritizes area-denial over single-target focus

Target Switching and Overkill Rules

- **Switch cooldown:** An ANPC will not switch targets more than once per 3 seconds unless the current target is destroyed or the ANPC is forced into CS-EVASIVE.
- **Overkill prevention:** When a target's hull integrity drops below 0.1 and two or more ANPCs are engaging it, the lowest-priority ANPC switches to the next-highest-priority target. This prevents "pile-on" behavior on near-dead targets.
- **Dead target release:** When a target enters CS-DESTROYED, all engaging ANPCs release the target and re-evaluate priorities within 0.5 seconds.

4.5 Morale and Attrition System

Every ANPC in Starfighter maintains a **morale** value as a float [0.0–1.0] that fluctuates based on combat events, ally status, and mission progress. Morale is stored as a position on the z=xy manifold surface, where x represents internal resolve (derived from Conscientiousness and Neuroticism) and y represents

external pressure (derived from the scenario vector's Social Pressure and Threat components).

Morale Modifiers

Event	Morale Delta	Scope
Ally destroyed	-0.15	All squadron members
Kill confirmed (self)	+0.10	Self only
Kill confirmed (ally)	+0.05	All squadron members
Player in close proximity (<500m)	+0.02/sec (max +0.10)	Allied ANPCs near player
Outnumbered (>2:1 ratio)	-0.10	All allied ANPCs (applied once on detection)
Reinforcements arrive	+0.15	All allied ANPCs
Mission objective completed	+0.10	All allied ANPCs
Mission objective failed	-0.20	All allied ANPCs
Friendly fire received	-0.10	Affected ANPC only
Enemy ace destroyed	+0.15	All allied ANPCs; -0.20 for all enemy ANPCs
Under sustained fire (5+ seconds)	-0.05/sec	Targeted ANPC only

Morale Thresholds and Behavioral Effects

Morale Range	Label	Behavioral Effect
> 0.7	High	Aggressive pursuit, confident callouts, willing to take risks, formation discipline strong
0.3 – 0.7	Medium	Standard behavior per personality vector; balanced aggression and caution
< 0.3	Low	Retreat evaluation triggered; defensive

Morale Range	Label	Behavioral Effect
		posture; surrender consideration (enemy only); panicked comms

Morale Contagion

Morale effects spread between squadron members via a contagion multiplier. When an ANPC's morale changes, nearby squadron members (within 1000m) receive a contagion delta equal to the original delta multiplied by 0.3. This creates cascade effects: a series of ally losses can trigger squadron-wide morale collapse, while a decisive kill can rally the group. The contagion does not cascade — only first-order propagation occurs to prevent runaway feedback loops.

Enemy morale: Enemy ANPCs use the same morale system. When enemy morale drops below 0.2, enemy pilots may surrender (broadcast surrender callout, cease fire, reduce speed to zero) or attempt to flee the combat area. Enemy aces have a morale floor of 0.15 — they will never surrender unless hull integrity is below 0.1 simultaneously.

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5. Starfighter Manifold Applications

The ButterflyFX™ manifold system is the computational backbone of all ANPC decision-making. Per Document #1 §6, the foundational surfaces $z=xy$ and $z=xy^2$ encode behavioral intensity as continuous geometry rather than discrete bytes. This section defines how these manifold surfaces are applied to combat scenarios in the Starfighter title.

5.1 Combat Manifold Mapping

Document #1 §7 defines the Scenario Vector as a 5-component tuple: [Threat, Opportunity, Ambiguity, Social Pressure, Time Pressure]. In Starfighter, each component maps to concrete combat parameters:

Scenario Vector Component	Starfighter Combat Mapping	Formula
Threat	Combined enemy combat power relative to friendly forces	(enemy_count × avg_enemy_combat_rating) / (friendly_count × avg_friendly_combat_rating), clamped to [0.0–1.0]
Opportunity	Target vulnerability and weapon match quality	(1.0 - target_shield_pct) × weapon_match_coefficient, where weapon_match_coefficient is 1.0 for ideal weapon/target pairing, 0.5 for neutral, 0.2 for poor match
Ambiguity	Information uncertainty in the combat space	sensor_interference_level + (unknown_contact_count / total_contact_count), clamped to [0.0–1.0]
Social Pressure	Chain of command pressure and player proximity influence	order_authority_weight × order_urgency + player_proximity_factor, where player_proximity_factor = max(0, 1.0 - (distance_to_player / 1000))
Time Pressure	Mission timer urgency and incoming threat ETA	max(mission_timer_urgency, 1.0 - (reinforcement_ETA_seconds / 120)), clamped to [0.0–1.0]

5.2 Manifold-Driven Combat Decisions

The two manifold surfaces serve distinct decision functions in combat:

z = xy (Linear Manifold): Used when behavioral response should scale proportionally with both input factors. In combat, when x = aggression impulse (derived from personality vector's Agreeableness-inverse and preferred_tactic) and y = tactical opportunity (Scenario Vector's Opportunity component), z determines

engagement intensity. A high z indicates full commitment to the engagement; a low z indicates restraint or disengagement.

$z = xy^2$ (Asymmetric Manifold): Used when one factor should have disproportionate influence — when escalation asymmetry matters. In combat, this surface models situations where small provocations cause disproportionate responses. When x = provocation level (incoming damage rate, insults from enemy ace, direct orders) and y = volatility (derived from Neuroticism and current morale), z determines **escalation magnitude**. A pilot with high volatility (y) will produce dramatically higher z values from the same provocation (x) compared to a stable pilot, because the y^2 term amplifies the effect.

WORKED EXAMPLE — Manifold Lookup for Wingman Under Ambush

Scenario: Wingman "Hotshot" (Marcus Chen) has high loyalty (disposition 0.75) but low aggression (preferred_tactic: balanced, Agreeableness: 0.7 → aggression impulse ~0.3). He receives a "Weapons Free" order during an ambush.

Step 1 — Compute Scenario Vector:
Ambush mission template (§10.1): [Threat=0.9, Opportunity=0.8, Ambiguity=0.1, Social_Pressure=0.3, Time_Pressure=0.9]

Step 2 — Compute x (aggression impulse):
Base aggression = 1.0 - Agreeableness = 1.0 - 0.7 = 0.3
Order modifier (Weapons Free on balanced pilot) = +0.25
Adjusted x = 0.3 + 0.25 = 0.55

Step 3 — Compute y (tactical opportunity):
Scenario Vector Opportunity = 0.8 (targets are vulnerable during ambush surprise window)
y = 0.8

Step 4 — Manifold lookup z = xy:
z = 0.55 × 0.8 = 0.44
Interpretation: Moderate engagement intensity — Hotshot will engage decisively but without reckless overcommitment. He fires lasers with intent but does not burn torpedo charges yet.

Step 5 — Check asymmetric surface $z = xy^2$ for escalation:

Hotshot's Neuroticism: 0.35 (low-moderate)

Provocation (ambush surprise): x_provocation = 0.7

Volatility: y_volatility = 0.35 (Neuroticism × morale_instability_factor)

z_escalation = 0.7 × $(0.35)^2$ = 0.7 × 0.1225 = 0.086

Interpretation: Very low escalation risk — Hotshot remains controlled. He does not panic or over-escalate.

Step 6 — Behavioral Output:

Engagement intensity z=0.44 maps to: Laser engagement, aggressive pursuit, formation break authorized.

Escalation z=0.086 maps to: No desperation maneuvers, no insubordination, stable comms tone.

Phrase selection: Combat Engagement pool, medium urgency, confident tone.

5.3 Weapon Selection via Manifold

Manifold output z from the linear surface z=xy directly influences weapon selection. The engagement intensity score is mapped to weapon tiers:

z Range (Engagement Intensity)	Weapon Selection	Behavioral Pattern
0.0 – 0.2 (Very Low)	Hold fire OR Scatter Shot defensive	Maintain distance, evasive posture, monitor only
0.2 – 0.4 (Low)	Lasers — probing shots	Tentative engagement, testing target response, ready to disengage
0.4 – 0.6 (Medium)	Lasers — sustained fire	Committed engagement, pursuit, standard combat maneuvers
0.6 – 0.8 (High)	Lasers + Scatter Shot combination	Aggressive pursuit, closing to scatter range, coordinated attacks
0.8 – 1.0 (Very High)	Proton Torpedoes or EMP Burst	Maximum commitment, heavy ordnance deployment, decisive strike

Pre-selection validation: Before committing to a weapon, the ANPC validates:

7. **Ammo check:** Does the ANPC have ammunition for the selected weapon? If not, fall back to next-lower tier.
8. **Range validation:** Is the target within the weapon's effective range profile? If not, close distance or select alternative.
9. **Friendly fire check:** Are allied units within the weapon's damage radius (EMP Burst) or flight path (Proton Torpedoes)? If so, delay or re-angle.
10. **Lock-on requirement:** Proton Torpedoes require 2.5 seconds of lock-on. If the ANPC cannot maintain lock (evasive target), fall back to Lasers.

5.4 Manifold Drift in Extended Combat

Per Document #1 §6.5, prolonged exposure to high-stimulus environments causes the manifold lookup point to drift, altering behavioral output even if the underlying personality vector is unchanged. In Starfighter, extended combat (duration > 120 seconds) introduces the following drift mechanics:

Drift Factor	Effect on Manifold Input	Rate
Combat Fatigue	Aggression impulse (x) decays toward 0.3; reaction times increase	-0.02/minute after 2 minutes of continuous combat
Adrenaline Spike	Neuroticism-derived volatility (y in $z=xy^2$) spikes by +0.2 on first engagement, then decays	Spike on first CS-ENGAGED entry; decays at -0.05/30 seconds
Tunnel Vision	Target priority bias increases toward current target; reduces re-evaluation frequency	+0.05 current-target bias per minute of continuous engagement with same target
Adaptation	Ambiguity component decreases as ANPC gathers combat data on enemy patterns	-0.03/minute (ANPC "learns" enemy behavior during combat)

Post-combat manifold reset: When an ANPC returns to CS-PATROL state after combat, all drift parameters reset over a 30-second cooldown period. The manifold lookup point returns to its pre-combat position. This prevents accumulated drift from carrying over into subsequent engagements and ensures each combat starts from a clean behavioral baseline.

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6. Starfighter Communication Protocol

6.1 Radio Communication Model

All ANPC communication in the Starfighter title occurs via simulated radio channels. There is no face-to-face dialogue during missions — all communication is voice-only, transmitted across radio frequencies. Between-mission interactions (hangar, briefing room, station) use face-to-face dialogue governed by Document #1 §11.

Communication Channels

Channel	ID	Scope	Who Hears	Usage
Squadron	CH-SQD	Local flight group	All ANPCs in the same squadron_id	Tactical coordination, formation orders, combat callouts
Fleet	CH-FLT	All friendly forces	All allied ANPCs in the mission	Fleet-wide orders, strategic updates, reinforcement announcements
Private	CH-PVT	Player only	Player only (not other ANPCs)	Personal comments, relationship-driven

Channel	ID	Scope	Who Hears	Usage
				dialogue, private warnings
Enemy (Intercepted)	CH-ENM	Enemy forces (intercepted)	Player and allied ANPCs with Scanner upgrade	Enemy chatter; provides intel; may be garbled or partial

Channel Selection Rules

- Combat callouts default to CH-SQD
- Orders from Fleet Admiral use CH-FLT
- Personal remarks (disposition-driven) use CH-PVT when disposition > 0.6 or < -0.3
- Enemy ace taunts are broadcast on CH-SQD (as intercepted enemy comms)
- Emergency callouts (Mayday) broadcast on both CH-SQD and CH-FLT simultaneously

Radio Discipline

ANPCs follow radio discipline protocols adapted from multi-service tactical brevity standards:

- **Callsign protocol:** All transmissions begin with the sender's callsign or "All ships" for broadcast. Example: "Hotshot, bogey at two o'clock high."
- **Brevity codes:** Standardized short-form callouts are used in combat (see §6.2). ANPCs do not use casual language during active combat states.
- **Acknowledgment:** Orders require acknowledgment. Wingmen respond with "{callsign}, copy" or "{callsign}, wilco." Refusals use "{callsign}, unable — {reason}."

- **Transmission ending:** Non-urgent transmissions end with "over." Transmissions requiring no response end with "out." These may be omitted during high-intensity combat to reduce chatter.

6.2 Combat Callout Taxonomy

The following standardized callouts are triggered by specific combat events. Each callout has defined trigger conditions and is selected from the phrase pool with urgency and tone modifiers applied.

Callout	Meaning	Trigger Condition	Channel
Bogey	Unidentified contact detected	New contact on sensors, not yet classified	CH-SQD
Bandit	Confirmed hostile contact	Contact positively identified as enemy	CH-SQD
Splash	Confirmed kill — enemy destroyed	Enemy ANPC enters CS-DESTROYED	CH-SQD
Fox One	Laser volley fired at target	ANPC fires sustained laser burst at designated target	CH-SQD
Fox Two	Scatter Shot fired	ANPC fires Scatter Shot at close-range target	CH-SQD
Fox Three	Proton Torpedo launched	ANPC launches Proton Torpedo (lock-on confirmed)	CH-SQD
Break Left / Break Right	Evasive maneuver instruction	Issued to ally under immediate threat from specific direction	CH-SQD
Tally	Visual confirmation on target	ANPC has line-of-sight on designated target	CH-SQD
No Joy	Lost visual on	Target moved	CH-SQD

Callout	Meaning	Trigger Condition	Channel
	target	behind obstacle or out of sensor range	
Winchester	All ammunition depleted	weapon_loadout ammo for all limited weapons reaches 0	CH-SQD
Bingo Fuel	Fuel at minimum return level	Fuel level reaches threshold requiring immediate return to base	CH-SQD + CH-FLT
Mayday	Critical emergency — imminent destruction	ANPC enters CS-DISABLED or hull integrity < 0.1	CH-SQD + CH-FLT
EMP Warning	EMP Burst deployment imminent	ANPC or ally activating EMP Burst — 3-second warning	CH-SQD
Tone	Torpedo lock-on acquired	Proton Torpedo lock-on sequence initiated (2.5s before launch)	CH-SQD
Reform	Return to assigned formation	All hostiles neutralized; Squadron Leader orders reformation	CH-SQD
Bug Out	Disengage and retreat	Squadron Leader orders retreat; morale below threshold	CH-SQD

6.3 Communication Frequency Rules

Communication frequency is governed by the **urgency scalar** derived from the Scenario Vector's combined Threat and Time Pressure components. The urgency scalar determines the minimum interval between consecutive transmissions from the same ANPC.

Urgency Scalar Range	Min Interval (seconds)	Label	Behavioral Note
0.0 – 0.3 (Low)	12 – 20	Casual	Patrol chatter, status checks, banter
0.3 – 0.6 (Medium)	6 – 12	Alert	Contact reports, threat assessment, tactical discussion
0.6 – 0.8 (High)	3 – 6	Combat	Rapid callouts, coordination, damage reports
0.8 – 1.0 (Critical)	1 – 3	Emergency	Overlapping comms, urgent warnings, Mayday priority

Chatter reduction: During sustained high-intensity combat (> 30 seconds at urgency > 0.8), non-essential communications are suppressed. Only mandatory callouts (see below) fire during chatter reduction.

Silence protocols: During stealth missions (ROE-HOLD active), all voluntary communication is suppressed. Only Mayday and Break callouts are permitted.

Mandatory callouts that always fire regardless of frequency limits or silence protocols:

- Mayday (CS-DISABLED entry)
- Break Left / Break Right (imminent collision or missile warning)
- Splash (first kill confirmation in an engagement — subsequent kills may be suppressed)
- EMP Warning (friendly EMP deployment)

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7. Starfighter Phrase Pools

Per Document #1 §8, phrase pools are organized in a three-tier hierarchy: Universal-Pool → Title-Pool → Character-Pool. This section defines the Title-Pool for Starfighter and provides Character-Pool examples for key ANPCs.

7.1 Title-Pool: Starfighter

The following phrase entries are available to all ANPCs in the Starfighter title. They extend the Universal-Pool phrases from Document #1 and may be overridden by Character-Pool entries. Each entry follows Document #1's Phrase Entry Format.

Combat Engagement Phrases

Phrase ID	Phrase Key	Phrase Template	Tone Vector	Urgency Range	Role Restrictions	Personality Constraints	Variable Slots
SF-CE-001	engage_target	"{callsign}, engaging {target_callsign}."	[0.6, 0.3, 0.7]	0.5 – 0.8	SF-WING, SF-SQLDR, SF-MERC	None	{callsign}, {target_callsign}
SF-CE-002	fox_three	"Fox Three! Torpedo away on {target_callsign}."	[0.7, 0.2, 0.8]	0.7 – 1.0	All combat roles	None	{target_callsign}
SF-CE-003	guns_guns	"Guns, guns, guns!"	[0.8, 0.1, 0.9]	0.6 – 0.9	All combat roles	Extraversion > 0.4	None
SF-CE-004	splash_confirmed	"Splash one {target_ship_class}! Good hit."	[0.7, 0.6, 0.5]	0.4 – 0.7	All combat roles	None	{target_ship_class}

Phrase ID	Phrase Key	Phrase Template	Tone Vector	Urgency Range	Role Restrictions	Personality Constraints	Variable Slots
SF-CE-005	tally_target	"Tally, I see him — {target_bearing}, {target_range}."	[0.5, 0.3, 0.6]	0.4 – 0.6	All combat roles	None	{target_bearing}, {target_range}
SF-CE-006	no_joy	"No joy on {target_callsign}. Lost visual."	[0.3, 0.4, 0.3]	0.4 – 0.7	All combat roles	None	{target_callsign}
SF-CE-007	weapons_free_ack	"{callsign}, weapons free — engaging at will."	[0.7, 0.2, 0.8]	0.5 – 0.7	SF-WING, SF-MERC	None	{callsign}
SF-CE-008	scatter_close	"Switching to scatter — closing range!"	[0.8, 0.1, 0.9]	0.6 – 0.9	All combat roles	Agreeableness < 0.6	None

Tactical Coordination Phrases

Phrase ID	Phrase Key	Phrase Template	Tone Vector	Urgency Range	Role Restrictions	Personality Constraints	Variable Slots
SF-TC-001	form_up	"All ships, form up — {formation_na	[0.5, 0.2, 0.7]	0.3 – 0.6	SF-SQLDR	None	{formation_name}

Phrase ID	Phrase Key	Phrase Template	Tone Vector	Urgency Range	Role Restrictions	Personality Constraints	Variable Slots
		me}."					
SF-TC-002	break_formation	"Breaking formation! Engage at will!"	[0.8, 0.1, 0.9]	0.7 – 1.0	SF-SQLDR, SF-WING	None	None
SF-TC-003	flank_order	"{target_callsign} is flanking — {wingman_callsign}, cut him off!"	[0.7, 0.2, 0.8]	0.6 – 0.8	SF-SQLDR	None	{target_callsign}, {wingman_callsign}
SF-TC-004	cover_me	"{callsign} to {wingman_callsign} — cover my six!"	[0.6, 0.3, 0.6]	0.5 – 0.8	SF-WING, SF-SQLDR	None	{callsign}, {wingman_callsign}
SF-TC-005	reform_order	"Hostiles clear. Reform on me, {formation_name}."	[0.4, 0.5, 0.5]	0.2 – 0.4	SF-SQLDR	None	{formation_name}
SF-TC-006	coordinated_strike	"All ships, coordinated strike on {target_callsign} — on my mark."	[0.7, 0.2, 0.8]	0.6 – 0.9	SF-SQLDR	None	{target_callsign}

Damage Report Phrases

Phrase ID	Phrase Key	Phrase Template	Tone Vector	Urgency Range	Role Restrictions	Personality Constraints	Variable Slots
SF-DR-001	shields_down	"{callsign}, shields are down! Hull taking hits."	[0.3, 0.7, 0.4]	0.7 – 0.9	All combat roles	None	{callsign}
SF-DR-002	hull_critical	"Hull critical — I'm not going to last much longer!"	[0.2, 0.9, 0.3]	0.8 – 1.0	All combat roles	Neuroticism > 0.3	None
SF-DR-003	system_failure	"{system_name} offline! Compensating."	[0.4, 0.6, 0.4]	0.6 – 0.8	All combat roles	None	{system_name}
SF-DR-004	winchester_call	"Winchester, winchester — I'm dry on {weapon_name}."	[0.3, 0.5, 0.4]	0.5 – 0.7	All combat roles	None	{weapon_name}
SF-DR-005	mayday_call	"Mayday, mayday! {callsign} going down — {position}!"	[0.1, 0.95, 0.2]	0.9 – 1.0	All combat roles	None	{callsign}, {position}

Morale/Banter Phrases

Phrase ID	Phrase Key	Phrase Template	Tone Vector	Urgency Range	Role Restrictions	Personality Constraints	Variable Slots
SF-MB-001	victory_cheer	"That's another one down! Nice shooting, {player_callsign}!"	[0.8, 0.1, 0.8]	0.2 – 0.5	SF-WING, SF-MERC	Extraversion > 0.5	{player_callsign}
SF-MB-002	taunt_enemy	"That all you've got? Come on!"	[0.9, 0.1, 0.9]	0.4 – 0.6	SF-WING, SF-EACE	Extraversion > 0.6, Agreeableness < 0.5	None
SF-MB-003	encourage_ally	"Stay with me, {wingman_callsign} — we've got this."	[0.6, 0.2, 0.6]	0.3 – 0.6	SF-SQLDR, SF-WING	Agreeableness > 0.5	{wingman_callsign}
SF-MB-004	patrol_idle	"Quiet out here. Almost too quiet."	[0.3, 0.1, 0.3]	0.0 – 0.2	SF-WING	None	None
SF-MB-005	post_combat_relief	"That was close. Everyone still in one piece?"	[0.5, 0.3, 0.4]	0.2 – 0.4	SF-SQLDR, SF-WING	Agreeableness > 0.4	None

Mission Communication Phrases

Phrase ID	Phrase Key	Phrase Template	Tone Vector	Urgency Range	Role Restrictions	Personality Constraints	Variable Slots
SF-MC-001	mission_ack	"{callsign}, copy mission parameters. Ready for launch."	[0.5, 0.1, 0.6]	0.1 – 0.3	All roles	None	{callsign}
SF-MC-002	objective_update	"New objective: {objective_text}. Adjusting course."	[0.5, 0.3, 0.5]	0.3 – 0.6	SF-CMDOP, SF-SQLDR	None	{objective_text}
SF-MC-003	extraction_request	"{callsign} requesting extraction at {nav_point}."	[0.3, 0.6, 0.4]	0.6 – 0.8	All combat roles	None	{callsign}, {nav_point}
SF-MC-004	mission_complete	"All objectives complete. RTB — good work, everyone."	[0.7, 0.1, 0.6]	0.1 – 0.3	SF-SQLDR, SF-CMDOP	None	None

Emergency Phrases

Phrase ID	Phrase Key	Phrase Template	Tone Vector	Urgency Range	Role Restrictions	Personality Constraints	Variable Slots
SF-EM-001	eject_warning	"Ejecting! {callsign} is punching out!"	[0.2, 0.95, 0.3]	0.9 – 1.0	All combat roles	None	{callsign}
SF-EM-002	emp_warning	"EMP hot in three — all friendlies clear the zone!"	[0.5, 0.8, 0.6]	0.8 – 1.0	All combat roles	None	None
SF-EM-003	collision_warning	"Break, break! Collision course — break {direction}!"	[0.2, 1.0, 0.5]	0.95 – 1.0	All roles	None	{direction}
SF-EM-004	reactor_critical	"Reactor going critical — get clear of my ship!"	[0.1, 1.0, 0.2]	0.95 – 1.0	All combat roles	None	None

7.2 Character-Pool Examples

The following three Character-Pools override Title-Pool defaults to give specific ANPCs unique voices. Per Document #1 §8, Character-Pool entries take highest priority in phrase selection.

"Frostbite" — Renna Volkov (Cold, Precise, Minimal)

Phrase ID	Overrides	Phrase Template	Tone Vector	Urgency Range
FB-CE-001	SF-CE-001	"Engaging."	[0.3, 0.1, 0.5]	0.5 – 0.8
FB-CE-004	SF-CE-004	"Target down."	[0.2, 0.1, 0.3]	0.3 – 0.5
FB-DR-001	SF-DR-001	"Shields gone."	[0.2, 0.4, 0.3]	0.6 – 0.8
FB-MB-001	SF-MB-001	"Adequate."	[0.2, 0.05, 0.2]	0.1 – 0.3
FB-EM-001	SF-EM-001	"Frostbite — ejecting."	[0.1, 0.5, 0.2]	0.9 – 1.0

"Hotshot" — Marcus Chen (Brash, Confident, Trash-Talking)

Phrase ID	Overrides	Phrase Template	Tone Vector	Urgency Range
HS-CE-001	SF-CE-001	"Hotshot's on him — watch this!"	[0.9, 0.1, 0.95]	0.5 – 0.8
HS-CE-004	SF-CE-004	"Boom! That's how it's done! Add another to the board!"	[0.95, 0.05, 0.95]	0.4 – 0.7
HS-DR-002	SF-DR-002	"Okay, okay — that one hurt. Still in it though!"	[0.6, 0.5, 0.6]	0.7 – 0.9
HS-MB-002	SF-MB-002	"You call that flying? My grandmother could dodge better than that!"	[0.95, 0.05, 0.95]	0.3 – 0.5
HS-EM-001	SF-EM-001	"No way — Hotshot doesn't eject! ...Okay, maybe this once — PUNCHING	[0.6, 0.9, 0.7]	0.9 – 1.0

Phrase ID	Overrides	Phrase Template	Tone Vector	Urgency Range
		OUT!"		

"Mother Hen" — Admiral Kira Osei (Calm Command, Warm but Firm)

Phrase ID	Overrides	Phrase Template	Tone Vector	Urgency Range
MH-TC-001	SF-TC-001	"All ships, form up — {formation_name}. Let's bring everyone home."	[0.5, 0.1, 0.5]	0.2 – 0.5
MH-MC-002	SF-MC-002	"Listen up. New objective: {objective_text}. Stay sharp and stay together."	[0.5, 0.3, 0.5]	0.3 – 0.6
MH-MC-004	SF-MC-004	"All objectives met. Outstanding work. Bring them home, Squadron Leader."	[0.6, 0.1, 0.5]	0.1 – 0.3
MH-MB-003	SF-MB-003	"Steady now, {wingman_callsign}. I've seen you fly through worse. You've got this."	[0.5, 0.2, 0.4]	0.3 – 0.6
MH-EM-003	SF-EM-003	"Collision alert! All ships, emergency break — {direction}, now!"	[0.4, 0.9, 0.6]	0.95 – 1.0

7.3 Enemy Phrase Pools

Enemy Ace Phrases

Phrase ID	Phrase Key	Phrase Template	Context
EA-TAUNT-001	ace_taunt_open	"So you're the one they talk about. Let's see if you're worth my time."	First engagement with player
EA-TAUNT-002	ace_taunt_mid	"Not bad — but not good enough."	Player hits enemy ace (hull > 0.5)
EA-TAUNT-003	ace_grudging	"You're better than I expected. This might actually be interesting."	Player damages enemy ace (hull < 0.5)
EA-SURR-001	ace_surrender	"...Enough. I yield. You've earned this."	Enemy ace morale < 0.15 AND hull < 0.1
EA-FINAL-001	ace_final	"Remember my name..."	Enemy ace enters CS-DESTROYED

Standard Enemy Pilot Phrases

Phrase ID	Phrase Key	Phrase Template	Context
EP-CONTACT-001	enemy_contact	"Contacts bearing {bearing}. Engaging."	Enemy detects player squadron
EP-HIT-001	enemy_hit	"Taking fire — requesting support."	Enemy pilot takes damage
EP-RETREAT-001	enemy_retreat	"Breaking off — too many of them!"	Enemy morale < 0.3

Enemy Fleet Commander Phrases

Phrase ID	Phrase Key	Phrase Template	Context
EFC-THREAT-001	fleet_threat	"You are trespassing in sovereign space. Withdraw or be destroyed."	Mission opening — diplomatic confrontation
EFC-ULTIM-001	fleet_ultimatum	"This is your final warning. All hostile craft will be eliminated."	Pre-combat escalation
EFC-RETREAT-001	fleet_retreat_order	"All units, fall back to rally point. We will return."	Enemy fleet morale collapse

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8. Starfighter Composite String Assembly

8.1 Combat String Templates

Per Document #1 §9, composite strings are assembled from four components: **[Opener + Core + Modifier + Closer]**. In Starfighter combat communications, these components map to radio transmission structure:

Component	Function	Examples
Opener	Identifies speaker and/or addressee	"{callsign}, " / "All ships, " / "Command, " / "{target_callsign}, "
Core	Primary message content	"I've got lock on {target_callsign}" / "taking heavy fire from {direction}" / "form up — {formation_name}"
Modifier	Additional context or status	", shields at {shield_pct}%" / ", {weapon_name} armed" / ", hull at {hull_pct}%"

Component	Function	Examples
Closer	Action declaration or protocol ending	", engaging!" / ", requesting backup!" / ", over." / ", out."

Assembly rule: Opener is always present. Core is always present. Modifier is optional — included when urgency > 0.5 or when the modifier conveys tactically relevant information. Closer is optional — omitted during high-intensity combat (urgency > 0.8) to reduce transmission length.

8.2 Worked Assembly Examples

Example 1: Wingman Reporting Contact

Step	Detail
Input Scenario Vector	[Threat=0.4, Opportunity=0.3, Ambiguity=0.7, Social_Pressure=0.2, Time_Pressure=0.1]
Manifold Lookup (z=xy)	x (aggression) = 0.35, y (opportunity) = 0.3 → z = 0.105 (very low intensity — report, do not engage)
Tone/Urgency Calc	Urgency = 0.4 (medium-low, contact not yet hostile). Tone: alert but not alarmed.
Phrase Selection	SF-CE-005 (tally_target) selected — matches "contact report" context, urgency within range
Variable Interpolation	{target_bearing} = "two o'clock high", {target_range} = "1200 meters"
Composite Assembly	**Opener:** "Hotshot, " + **Core:** "bogey at two o'clock high, 1200 meters" + **Modifier:** (omitted — low urgency) + **Closer:** ", over."
Final String	**"Hotshot, bogey at two o'clock high, 1200 meters, over."**

Example 2: Squadron Leader Issuing Formation Change

Step	Detail
Input Scenario Vector	[Threat=0.6, Opportunity=0.5, Ambiguity=0.3, Social_Pressure=0.5, Time_Pressure=0.6]
Manifold Lookup (z=xy)	x (command authority) = 0.7, y (tactical need) = 0.6 → z = 0.42 (moderate intensity — tactical repositioning)
Tone/Urgency Calc	Urgency = 0.55 (medium). Tone: authoritative, measured.
Phrase Selection	SF-TC-001 (form_up) selected — squadron leader role, formation context
Variable Interpolation	{formation_name} = "Diamond"
Composite Assembly	**Opener:** "All ships, " + **Core:** "form up — Diamond" + **Modifier:** ", we're entering the debris field" + **Closer:** ", over."
Final String	**"All ships, form up — Diamond, we're entering the debris field, over."**

Example 3: Command Base Delivering Mission Update

Step	Detail
Input Scenario Vector	[Threat=0.3, Opportunity=0.4, Ambiguity=0.5, Social_Pressure=0.6, Time_Pressure=0.7]
Manifold Lookup (z=xy)	x (information urgency) = 0.6, y (time pressure) = 0.7 → z = 0.42 (moderate priority delivery)
Tone/Urgency Calc	Urgency = 0.6 (medium-high). Tone: calm, precise, data-rich.
Phrase Selection	SF-MC-002 (objective_update) selected — CMDOP role, mission context
Variable Interpolation	{objective_text} = "escort convoy to nav point Bravo-Seven"

Step	Detail
Composite Assembly	**Opener:** "Lighthouse to all ships, " + **Core:** "new objective — escort convoy to nav point Bravo-Seven" + **Modifier:** ", enemy reinforcements inbound ETA four minutes" + **Closer:** ", out."
Final String	**"Lighthouse to all ships, new objective — escort convoy to nav point Bravo-Seven, enemy reinforcements inbound ETA four minutes, out."**

Example 4: Enemy Ace Taunting During Dogfight

Step	Detail
Input Scenario Vector	[Threat=0.5, Opportunity=0.7, Ambiguity=0.1, Social_Pressure=0.1, Time_Pressure=0.3]
Manifold Lookup ($z=xy^2$)	x (provocation — player hit the ace) = 0.6, y (volatility from ego) = 0.7 → z = 0.6 × 0.49 = 0.294 (moderate escalation — taunting, not enraged)
Tone/Urgency Calc	Urgency = 0.45 (medium). Tone: confident, mocking, personal.
Phrase Selection	EA-TAUNT-002 (ace_taunt_mid) selected — context: player hit ace, hull > 0.5
Variable Interpolation	None required (no variable slots)
Composite Assembly	**Opener:** (none — enemy ace breaks protocol for dramatic effect) + **Core:** "Not bad — but not good enough." + **Modifier:** (omitted) + **Closer:** (omitted — taunt needs no closing)
Final String	**"Not bad — but not good enough."**

Example 5: Damaged Wingman Calling for Retreat

Step	Detail
Input Scenario Vector	[Threat=0.9, Opportunity=0.1, Ambiguity=0.1, Social_Pressure=0.3, Time_Pressure=0.8]
Manifold Lookup (z=xy)	x (aggression, suppressed by damage) = 0.15, y (opportunity) = 0.1 → z = 0.015 (near-zero intensity — retreat)
Manifold Lookup ($z=xy^2$)	x (sustained fire provocation) = 0.8, y (Neuroticism 0.45 + morale penalty) = 0.6 → z = 0.8 × 0.36 = 0.288 (moderate panic)
Tone/Urgency Calc	Urgency = 0.85 (high). Tone: stressed, desperate, requesting help.
Phrase Selection	SF-DR-002 (hull_critical) + SF-MC-003 (extraction_request) chained
Variable Interpolation	{callsign} = "Frostbite", {nav_point} = "Rally Point Alpha"
Composite Assembly	**Opener:** "Frostbite — " + **Core:** "hull critical, I'm pulling out" + **Modifier:** ", shields gone, engines at forty percent" + **Closer:** ", heading to Rally Point Alpha!"
Final String	**"Frostbite — hull critical, I'm pulling out, shields gone, engines at forty percent, heading to Rally Point Alpha!"**

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9. Starfighter Interaction Rules

9.1 Player-to-ANPC Combat Interactions

Command Interface: The player issues orders to assigned wingmen through a radial command menu. Available commands include:

- **Target Designation:** Player marks a target; wingmen prioritize that target (+0.4 priority bias)

- **Formation Change:** Player selects from available formations; wingmen reposition
- **ROE Change:** Player sets Weapons Free / Weapons Tight / Weapons Hold for assigned wingmen
- **Attack My Target:** All wingmen focus fire on player's current target
- **Cover Me:** Wingmen adopt defensive posture around player, engaging threats to player
- **Break and Engage:** Formation breaks; all wingmen engage targets freely
- **Retreat:** All wingmen disengage and head to nearest safe zone

ANPC Compliance Model: An ANPC's willingness to comply with a player order is calculated as:

compliance_score = disposition × (1.0 - order_danger_level) × morale

If compliance_score > 0.3, the ANPC complies. If compliance_score is between 0.15 and 0.3, the ANPC complies with a verbal objection. If compliance_score < 0.15, the ANPC refuses and states the reason.

Refusal dialogue examples:

- "Negative, {player_callsign} — that's a suicide run. I'm not throwing my ship away." (order_danger_level > 0.8, low morale)
- "{player_callsign}, that contradicts standing orders from command. Unable to comply." (order conflicts with higher ROE)
- "I hear you, but I've got three on my tail — give me a second!" (ANPC in CS-EVASIVE, cannot reposition)

Praise/Criticism System: Player combat performance affects ANPC disposition over time:

- Player achieves kills efficiently: wingmen disposition +0.02 per kill (respect for competence)
- Player takes excessive damage: wingmen disposition -0.01 per damage event (concern / frustration)

- Player issues orders that lead to wingman survival: disposition +0.05 per mission (trust building)
- Player issues orders that lead to wingman destruction: disposition -0.15 per death (trust erosion)

9.2 ANPC-to-ANPC Combat Interactions

- **Wingman-to-wingman coordination:** Two wingmen engaging the same target will automatically deconflict — one attacks from above, the other from below or behind. They announce their approach vector: "I'll take him high — you go low."
- **Squadron leader tactical orders:** The squadron leader evaluates the combat space every 5 seconds and issues tactical directives to wingmen: target assignments, flanking orders, and retreat commands. These are processed through the same compliance model as player orders.
- **Cross-squadron communication:** In fleet battles with multiple squadrons, squadron leaders coordinate via CH-FLT. They deconflict engagement zones and share intel on high-value targets.
- **Enemy-to-enemy coordination:** Enemy pilots in the same group coordinate flanking approaches. When one enemy pilot engages the player head-on, another attempts to attack from the flank. Enemy aces may order subordinate pilots to herd the player into a kill zone.

9.3 Between-Mission Interactions

Between missions, ANPCs are available for face-to-face dialogue in hangar, briefing, and station environments. These interactions follow Document #1 §11 for general dialogue rules, with the following Starfighter-specific extensions:

- **Hangar interactions:** Wingmen and mechanics are present in the hangar. Wingmen discuss the previous mission, share personal backstory fragments (gated by disposition level), and comment on ship upgrades. Mechanics report ship condition and offer upgrade options.

- **Pre-mission briefings:** The Squadron Leader or Command Base Operator delivers the briefing. Wingmen may interject with questions, concerns, or enthusiasm based on personality and mission danger level.
- **Post-mission debriefing:** ANPCs comment on mission performance. High-performing players receive praise; poor performance triggers constructive criticism from the Squadron Leader and concern from wingmen. Kill counts are updated and referenced in dialogue.
- **Relationship progression:** As disposition increases, ANPCs unlock deeper dialogue: personal backstory, motivations, fears, and relationship-specific combat bonuses (coordinated attack phrases, unique callouts, +0.05 mutual combat_rating boost when flying together).

9.4 Disposition Shifts from Combat Events

The following table defines how combat events shift ANPC disposition toward the player. These deltas apply immediately and persist across sessions (per Document #1 §5).

Combat Event	Disposition Delta	Conditions / Notes
Player saves ANPC from destruction (kills attacker targeting ANPC with hull < 0.3)	+0.15 (large positive)	ANPC must be in CS-EVASIVE or CS-DAMAGED when saved
Player ignores ANPC distress call (Mayday unanswered for 10+ seconds)	-0.12 (large negative)	Only triggers if player was within 1500m and had no current engagement
Player assigns ANPC to suicide mission (order_danger_level > 0.8)	-0.08 (medium negative)	+0.03 instead if ANPC has Conscientiousness > 0.8 and preferred_tactic = aggressive (fanatic)
ANPC and player achieve kill together (both dealing damage to same target)	+0.04 (small positive)	Cooperative kill; builds camaraderie
Extended combat together without losses (> 3 minutes, no allied	+0.03 (small positive)	Applied once at mission end

Combat Event	Disposition Delta	Conditions / Notes
deaths)		
Player friendly-fires ANPC	-0.10 (large negative)	Scales with damage dealt; -0.15 if ANPC hull drops below 0.5
Player completes mission with all wingmen alive	+0.05 (positive, all wingmen)	Applied to all surviving wingmen at debrief
Player abandons wingman during retreat (retreats while wingman still engaged)	-0.06 (medium negative)	Only triggers if ANPC has Agreeableness > 0.5
Player follows ANPC's tactical suggestion (acts on advice within 5 seconds)	+0.03 (small positive)	Validates ANPC input; builds mutual respect
Player repeatedly overrides ANPC (3+ overrides in one mission)	-0.04 (small negative)	ANPC feels disrespected / marginalized

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10. Starfighter Scenario Vectors

10.1 Mission Type Scenario Templates

Each mission type in Starfighter has a pre-defined scenario vector template that is loaded at mission start and serves as the baseline for dynamic modification. The vector format is [Threat, Opportunity, Ambiguity, Social Pressure, Time Pressure].

Mission Type	Threat	Opportunity	Ambiguity	Social Pressure	Time Pressure	Description
Patrol	0.2	0.3	0.5	0.2	0.1	Low threat, moderate uncertainty, relaxed timing

Mission Type	Threat	Opportunity	Ambiguity	Social Pressure	Time Pressure	Description
Escort	0.4	0.2	0.3	0.6	0.5	Moderate threat, high responsibility for convoy, steady pace
Assault	0.8	0.7	0.2	0.4	0.7	High threat and opportunity, clear targets, time-critical
Reconnaissance	0.3	0.5	0.8	0.3	0.4	Low threat, high ambiguity, intelligence-gathering focus
Defense	0.7	0.3	0.4	0.7	0.8	High threat and social pressure, defending a fixed position
Ambush	0.9	0.8	0.1	0.3	0.9	Extreme threat and opportunity, surprise engagement, time-critical

10.2 Dynamic Scenario Vector Modifiers

In-game events modify the scenario vector in real-time. Modifiers are additive and applied immediately upon event detection. Multiple modifiers can stack within the same frame.

In-Game Event	Threat Δ	Opportunity Δ	Ambiguity Δ	Social Pressure Δ	Time Pressure Δ
Ambush detected (enemy surprise attack)	+0.4	-0.1	-0.3	+0.1	+0.3
Allied ANPC destroyed	+0.2	-0.05	0.0	+0.1	0.0
New intel received (sensor update)	0.0	+0.1	-0.2	0.0	0.0
Allied reinforcements arrive	-0.2	+0.1	-0.1	-0.05	-0.1
Enemy reinforcements detected	+0.3	-0.1	+0.1	+0.1	+0.2
Mission objective completed	-0.1	0.0	-0.1	-0.1	-0.2
Mission objective failed	+0.1	-0.2	+0.1	+0.2	+0.1
Civilian vessel enters combat zone	0.0	-0.1	+0.2	+0.3	+0.1
EMP burst detonated (friendly or enemy)	-0.1	+0.2	+0.1	0.0	0.0

In-Game Event	Threat Δ	Opportunity Δ	Ambiguity Δ	Social Pressure Δ	Time Pressure Δ
Enemy ace enters combat	+0.2	+0.1	-0.1	+0.1	0.0

10.3 Scenario Vector Boundaries

To ensure behavioral stability and prevent jarring state transitions, the following boundary rules apply to all scenario vector operations:

- **Clamping:** All five components are hard-clamped to [0.0–1.0] after every modification. Values exceeding 1.0 are clamped to 1.0; values below 0.0 are clamped to 0.0.
- **Rate-of-change limit:** No single component may change by more than 0.4 per second. If an event would cause a larger delta, the change is applied over multiple frames using linear interpolation. This prevents instantaneous behavioral whiplash.
- **Smoothing algorithm:** Scenario vector transitions use exponential moving average (EMA) smoothing with a decay factor of 0.85. For each frame: $V_new = V_current \times 0.85 + V_target \times 0.15$. This produces smooth, natural-feeling behavioral transitions.
- **Stability threshold:** If the scenario vector has not changed by more than 0.02 in any component for 10 seconds, the ANPC enters a "behavioral steady state" that reduces manifold recalculation frequency from per-frame to every 0.5 seconds, saving CPU cycles.

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11. Developer Implementation Notes — Starfighter Specific

11.1 ANPC Spawning and Initialization

Spawn point system: Allied ANPCs spawn at pre-defined spawn points linked to formation positions. The player's starting hangar or carrier defines the spawn origin; wingmen spawn in their assigned formation positions relative to the player. Enemy ANPCs spawn at mission-scripted waypoints or procedurally generated intercept vectors.

Mission-scripted vs. procedural ANPCs:

- **Scripted ANPCs:** Named characters (Hotshot, Frostbite, etc.) with full identity schemas defined by the designer. All identity fields, personality vectors, backstory, and phrase pool assignments are hand-authored.
- **Procedural ANPCs:** Generic enemy pilots and reinforcements generated at runtime. Identity fields are populated from randomized ranges defined per role (§2.2). Callsigns are generated from a callsign pool. Personality vectors are randomized within role constraints.

Initial manifold calibration: On spawn, each ANPC's manifold lookup point is initialized at the intersection of their personality vector and the mission's starting scenario vector. This initial calibration takes 1 frame and produces the ANPC's behavioral baseline for the mission.

11.2 Performance Budget

Resource	Budget	Notes
Maximum concurrent ANPCs per scene	32	Includes all allied, enemy, and neutral ANPCs. Drone swarms count as 1 ANPC (the controller).
Manifold calculation frequency	Every 0.25 seconds (4 Hz) during combat; every	Per-frame calculation is not required. The EMA

Resource	Budget	Notes
	1.0 second during patrol	smoothing (§10.3) handles inter-frame interpolation.
Phrase pool memory footprint	~2 KB per ANPC (character pool) + 8 KB shared (title pool)	Phrase templates stored as string references, not duplicated instances.
Behavior tree evaluation	Every 0.1 seconds (10 Hz)	Behavior tree nodes query manifold output rather than recalculating from scratch.
Communication queue	Max 3 queued transmissions per ANPC	Oldest queued transmission is discarded if queue is full (except Mayday).

LOD system for distant ANPCs: ANPCs beyond 3000m from the player use a simplified behavioral model:

- Manifold calculations reduced to every 2.0 seconds
- Phrase pool disabled (no audible dialogue)
- Combat state transitions still process but at reduced frequency
- Full behavioral model restored when ANPC comes within 2000m of the player

11.3 Combat AI Tuning Parameters

The following parameters are exposed to designers for balancing and are accessible in the game editor without code changes:

Parameter	Type	Default	Range	Effect
aggression_multiplier	float	1.0	[0.1 – 3.0]	Scales the aggression impulse (x) before manifold lookup
accuracy_curve	float	0.6	[0.0 – 1.0]	Base hit probability; modified by

Parameter	Type	Default	Range	Effect
				combat_rating and distance
reaction_time	float (seconds)	0.5	[0.1 – 2.0]	Delay between threat detection and response initiation
formation_tightness	float	0.5	[0.0 – 1.0]	Spacing between formation positions; 0 = wide, 1 = parade-tight
evasion_skill	float	0.5	[0.0 – 1.0]	Effectiveness of evasive maneuvers; affects dodge probability
morale_sensitivity	float	1.0	[0.5 – 2.0]	Multiplier applied to all morale deltas
comms_frequency_multiplier	float	1.0	[0.5 – 2.0]	Adjusts minimum comm interval; higher = more chatter

Difficulty scaling: These tuning parameters vary by difficulty level, as detailed in Appendix E.

Playtest metrics to measure:

- Average player death count per mission (target: Easy = 0.3, Normal = 1.2, Hard = 2.5, Veteran = 4+)
- Average wingman survival rate per mission (target: > 80% on Easy, > 50% on Normal, > 30% on Hard)
- Player engagement duration per combat encounter (target: 45–120 seconds)

- ANPC communication volume per mission (target: 25–60 transmissions, no overlapping audio)
- Morale-driven retreats per mission (target: 0–1 on Easy, 1–3 on Normal, 2–5 on Hard)

11.4 Save/Load Considerations

The following ANPC state **persists** across save/load cycles:

- All Identity Schema fields (base and extended) including disposition, kill_count, flight_hours, loyalty_score, combat_rating
- Personality vector values (all 5 axes)
- Relationship disposition values toward the player and other named ANPCs
- Squadron assignment and callsign
- Narrative flags (backstory unlocks, relationship milestones, insubordination history)

The following state **resets** on load:

- Manifold drift parameters — fresh calibration is performed on load (ensures clean behavioral baseline)
- Phrase cooldown timers — all phrase cooldowns reset to 0 on load
- Combat state — all ANPCs reset to CS-PATROL on mission load
- Morale — resets to base value derived from personality vector on load (combat morale is transient)
- Communication queue — cleared on load
- Scenario vector — resets to mission template values on load

11.5 Multiplayer Considerations

Co-op mode: In co-op scenarios, ANPCs are assigned to specific players using a primary_player field in the identity schema. Each ANPC follows the orders of their assigned player and tracks disposition separately for each human player.

ANPC assignment: At mission start, available wingmen are distributed equally among human players. The host player has priority on named (scripted) ANPCs; additional players receive procedurally generated wingmen unless the mission script overrides.

Conflict resolution: When multiple players issue contradictory orders to the same ANPC (possible only for shared squadron-level ANPCs like the Squadron Leader), the following priority applies:

11. The player with the highest mission authority (host or designated flight lead) takes priority
12. If authority is equal, the ANPC follows the order from the player with the highest disposition score toward the ANPC
13. If still tied, the ANPC follows the most recent order
14. The ANPC announces the conflict: "Copy, {player_callsign} — following your lead on this one."

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12. Glossary — Starfighter Extensions

The following terms extend Document #1's Glossary with Starfighter-specific definitions. For base ANPC terms, refer to Document #1 §13.

Term	Definition
Attrition	Cumulative loss of combat capability through sustained engagement; modeled via morale decay and manifold drift
Bandit	Positively identified hostile contact; engagement permitted under ROE-TIGHT or ROE-FREE
Bogey	Unidentified contact detected on sensors; requires classification before engagement under ROE-TIGHT

Term	Definition
Combat Rating	Float [0.0–1.0] representing an ANPC's overall combat effectiveness; affects accuracy, reaction time, and tactical sophistication
Combat State	One of eight defined behavioral states (Patrol, Alert, Engaged, Evasive, Damaged, Retreating, Disabled, Destroyed) governing ANPC priorities
Damage Threshold	Hull integrity fraction at which an ANPC begins behavioral degradation (retreat evaluation, desperation, surrender consideration)
EMP Window	The 6–10 second vulnerability period after an EMP Burst detonation when affected ships have disabled shields and systems
Engagement Envelope	The spatial and parametric range within which an ANPC will initiate or continue combat with a target; defined by weapon range, combat_rating, and ROE
Flight Hours	Simulated flight experience accumulated by an ANPC; affects skill curves, dialogue references, and perceived seniority
Formation	Spatial arrangement of ships in a flight group; governs positioning, mutual support zones, and communication patterns
Kill Count	Total confirmed enemy eliminations credited to an ANPC; affects reputation, dialogue tone, and morale bonuses
Loyalty Score	Float [0.0–1.0] for mercenary and allegiance-fluid roles; below 0.3 triggers side-switch evaluation
Morale	Float [0.0–1.0] representing an ANPC's will to continue fighting; influenced by combat events, ally status, and manifold position
ROE (Rules of Engagement)	Command-level directive governing when ANPCs may fire weapons; three levels: Weapons Free, Weapons Tight, Weapons Hold

Term	Definition
Ship Class	Category of starfighter an ANPC pilots; determines behavioral modifiers: Interceptor, Bomber, Heavy Fighter, Scout, Capital Ship
Sortie	A single mission flight from launch to landing; used as a unit of measurement for flight_hours and statistical tracking
Splash	Confirmed destruction of an enemy target; triggers morale bonus and kill_count increment
Squadron	Organizational unit of 2–8 ships operating under a single Squadron Leader; identified by squadron_id and squadron_name
Torpedo Lock	2.5-second lock-on sequence required before Proton Torpedo launch; announced via "Tone" callout
Weapons Free	ROE level permitting engagement of any hostile contact without authorization; maximum combat autonomy
Weapons Hold	ROE level prohibiting all weapons fire except in immediate self-defense; used during stealth and diplomatic missions
Weapons Tight	ROE level permitting engagement of only positively identified hostile targets or command-designated targets

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13. Document Index — Starfighter

Topic	Section
Accuracy curve (tuning parameter)	§11.3
Adrenaline spike (manifold drift)	§5.4
Aggression multiplier	§11.3
Alert state (CS-ALERT)	§4.2

Topic	Section
Ambush scenario vector template	§10.1
ANPC lifecycle	§15.5
ANPC spawning	§11.1
Assault scenario vector template	§10.1
Asymmetric manifold ($z=xy^2$)	§5.2
Attrition system	§4.5
Between-mission interactions	§9.3
Bogey (callout)	§6.2
Bomber (ship class)	§3.2
Break Left/Right (callout)	§6.2
Callsign protocol	§6.1
Capital Ship (ship class)	§3.2
Chain of command	§2.3
Character-pool examples	§7.2
Civilian Contact (SF-CIV)	§2.1, §2.2
Clamping rules (scenario vector)	§10.3
Combat fatigue	§5.4
Combat rating	§3.1
Combat state machine	§4.2, §15.2
Command Base Operator (SF-CMDOP)	§2.1, §2.2
Communication channels	§6.1
Communication frequency rules	§6.3
Compliance model (player orders)	§9.1
Composite string assembly	§8
Damage threshold	§3.1
Defense scenario vector template	§10.1
Diamond formation	§4.3
Difficulty scaling	§11.3, Appendix E
Disposition shifts (combat events)	§9.4

Topic	Section
Drone Controller (SF-DRONE)	§2.1, §2.2
Dynamic scenario vector modifiers	§10.2
EMP Burst (weapon)	§3.3
Enemy Ace (SF-EACE)	§2.1, §2.2
Enemy phrase pools	§7.3
Enemy Pilot (SF-EPLT)	§2.1, §2.2
Escort scenario vector template	§10.1
Evasive state (CS-EVASIVE)	§4.2
Fleet Admiral (SF-FADM)	§2.1, §2.2
Flight hours	§3.1
Formation directives	§4.3
Formation tightness (tuning)	§11.3
Frostbite (character pool)	§7.2
Friendly fire (disposition impact)	§9.4
Hangar interactions	§9.3
Heavy Fighter (ship class)	§3.2
Hotshot (character pool / sample ANPC)	§7.2, Appendix A
Inheritance model (Doc #1 to Doc #2)	§1.2
Insubordination mechanics	§2.3
Interceptor (ship class)	§3.2
Kill count	§3.1
Lasers (weapon)	§3.3
Lighthouse (sample ANPC)	Appendix C
Linear manifold (z=xy)	§5.2
LOD system	§11.2
Loyalty score	§3.1
Manifold drift	§5.4
Mayday (callout)	§6.2
Mercenary (SF-MERC)	§2.1, §2.2

Topic	Section
Mission script integration	Appendix D
Morale contagion	§4.5
Morale system	§4.5
Mother Hen (character pool)	§7.2
Multiplayer considerations	§11.5
Nightshade (sample ANPC)	Appendix B
Order refusal	§2.3, §9.1
Overkill prevention	§4.4
Patrol scenario vector template	§10.1
Performance budget	§11.2
Phrase pool architecture	§7
Playtest metrics	§11.3
Post-combat manifold reset	§5.4
Proton Torpedoes (weapon)	§3.3
Radio discipline	§6.1
Rate-of-change limits (scenario vector)	§10.3
Reaction time (tuning parameter)	§11.3
Reconnaissance scenario vector template	§10.1
Refusal dialogue	§9.1
Retreat state (CS-RETREAT)	§4.2
ROE (Rules of Engagement)	§4.1
Role taxonomy	§2.1
Save/load considerations	§11.4
Scatter Shot (weapon)	§3.3
Scenario vector boundaries	§10.3
Scout (ship class)	§3.2
Ship class behavioral modifiers	§3.2
Ship Mechanic (SF-MECH)	§2.1, §2.2
Smoothing algorithm (scenario vector)	§10.3

Topic	Section
Splash (callout)	§6.2
Squadron Leader (SF-SQLDR)	§2.1, §2.2
State transitions (combat)	§4.2
"Store Geometry Not Bytes"	§1.1, §3.1, §5
Target priority system	§4.4
Tone (torpedo lock callout)	§6.2
Weapon selection via manifold	§5.3
Weapon system taxonomy	§3.3
Winchester (callout)	§6.2
Wingman (SF-WING)	§2.1, §2.2

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14. References

15. **Document #1 — General ANPC Guideline Document, ButterflyFX™ ANPC Design Suite, v1.0.** Kenneth Bingham. Defines universal ANPC architecture, identity schema, personality vectors, disposition system, manifold surface mathematics ($z=xy$, $z=xy^2$), phrase pool architecture, composite string assembly, and simulation-awareness levels. All Starfighter ANPC rules inherit from and must not contradict this document.
16. **ButterflyFX™ Manifold Surface Specification.** Kenneth Bingham. Formal mathematical specification of the $z=xy$ and $z=xy^2$ manifold surfaces, including coordinate space definitions, interpolation methods, boundary behavior, and the "Store Geometry Not Bytes" encoding philosophy.
17. **Multi-Service Brevity Codes (ATP 1-02.1 / MCRP 3-30B.1 / NTTP 6-02.1 / AFTTP 3-2.5).** ALSSA Center, January 2025 edition. Standardized tactical brevity codes for multi-service operations. Adapted for Starfighter's radio communication model with space-combat-appropriate modifications to terminology.

18. **Finite State Machines and Behavior Trees for Game AI.** Industry-standard game AI design patterns for NPC state management, goal-driven autonomy, and behavioral modularity. Informs the combat state machine (§4.2) and behavioral evaluation architecture.
19. **Formation Flying Algorithms for Real-Time Strategy Games.** Spatial arrangement algorithms for unit formation, dynamic reformation, and collision avoidance. Informs Starfighter's formation directive system (§4.3).
20. **Military Communication Protocols Adapted for Game Use.** Game design reference for adapting real-world radio discipline, channel management, and callsign conventions into interactive entertainment. Informs §6.

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15. Relational Diagrams

15.1 Chain of Command Flow

```
ALLIED FORCES — ORDER PROPAGATION  +-------------------+ |  FLEET ADMIRAL
|  Sets campaign-level ROE, strategic objectives |    (SF-FADM)     |  Issues orders
via CH-FLT +--------+---------+          |          | Campaign Directives (cannot be
overridden by subordinates)          | +--------v---------+     +--------------------+ |
SQUADRON LEADER  |

<--- data-id="3089">|  COMMAND BASE OPS   | |    (SF-SQLDR)     |     |    (SF-
CMDOP)       | |  May TIGHTEN ROE  |     |  Advisory only —    | |  (never loosen)   |
|  intel, extraction  | +--------+---------+     +--------------------+          |          | Tactical
Orders (wingmen may refuse if compliance_score

< 0.15)          |     +----+----+--------------------+     |         |                  | +---v---+ +---v---+
```

```
+------v------+ |WINGMAN| |WINGMAN|        |DRONE CTRL   | |(SF-WING)|(SF-WING)       |(SF-DRONE)   | +-------+ +-------+        +------+------+
|                          +------v------+                          |DRONE SWARM  |
|(automated)  |                          +-------------+  PLAYER AUTHORITY:   - Player holds Squadron Leader authority when assigned command   - Player orders propagate to directly assigned wingmen only   - Player ROE cannot exceed Fleet Admiral's ROE ceiling  NEUTRAL / OUTSIDE CHAIN:   SF-CIV  (Civilian Contact) — dialogue only, no orders   SF-MERC (Mercenary)        — follows player orders when loyalty > 0.5   SF-MECH (Ship Mechanic)    — reports to base command, not flight chain
```

15.2 Combat State Machine Diagram

```
+-------------+                    |   PATROL    |                    |  CS-PATROL  |
+------+------+                          |              Contact detected (2000m)
|                    +------v------+            +--------|    ALERT    |--------+            |        |  CS-ALERT  |        |            |        +------+------+        |      No contact            |          Contact = friendly      for 15s              |          (return to PATROL)      (to PATROL)          |
Bandit + ROE permits                          |                    +------v------+            +--------|
ENGAGED   |--------+--------+            |        |  CS-ENGAGED |        |        |            |
+------+------+        |        |            |              |              |        |      All hostiles     3+ hits/2s      Hull
< dmg_   |      cleared          (evasion        threshold      |      (to PATROL)      trigger)
(damage)       |                          |              |        Hull = 0                    +------v------+
+-----v-----+  (DESTROYED)            +--------|   EVASIVE   | |  DAMAGED   |            |
| CS-EVASIVE  |  | CS-DAMAGED |----+            |        +------+------+  +-----+------+   |
```

```
|             |             |           |     Evaded +        Morale
< 0.3   Morale
< 0.3     |     morale > 0.4     or evasion     or hull
< 0.15   |     (to ENGAGED)     fail 5s            |           |                        |             |
Hull
< 0.05                    +------v--------------v---+     or systems                    |
RETREATING        |     fail                  |     CS-RETREAT        |       |
+------+------+-----------+  +-----v-----+                        |     |            | DISABLED   |
Cornered +   Hull = 0.05       |CS-DISABLED |                 morale>0.2   (to
DISABLED)     +-----+------+                 (to ENGAGED)                      |
Damage or                                          self-destruct
|                                      +------v------+                                    |
DESTROYED   |                                      | CS-DESTROYED |
| (terminal)   |                                     +--------------+  NOTE: Any State -->
DESTROYED when hull = 0.0
```

15.3 Weapon Selection Decision Tree

```
MANIFOLD OUTPUT (z = engagement intensity)                |        +-------v-------+
| z >= 0.8?     |        | (Very High)   |        +---+-------+---+         Yes|       |No            |
|     +-------v---+   +-------v-------+     |Check: Has |   | z >= 0.6?     |     |torpedoes  |   |
(High)        |     |or EMP?    |   +---+-------+---+     +---+---+---+    Yes|       |No      Yes|
|No         |       |        |   |   +-------v---+   +-------v-------+        |   |   |Lasers +   |   | z >=
0.4?     |        |   |   |Scatter    |   | (Medium)      |        |   |   |combo      |   +---+-------+---
+        |   |   +-----------+    Yes|       |No         |   |                    |       |        |   +--->
Fallback:  +------v---+   +-------v-------+        |        Lasers +   |Lasers -  |   | z >= 0.2?
```

```
|        |        Scatter    |sustained |   | (Low)        |        |                |fire      |   +---
+-------+---+        |                +----------+    Yes|      |No    +----v--------+
|      |    |Target in    |                +-------v--+ +--v----------+    |range?       |
|Lasers -  | |Hold fire OR |    +---+-----+---+                |probing   | |Scatter Shot |
Yes|     |No                    |shots     | |defensive    |        |     |                    +----------
+ +-------------+    +---v---+ +--v--------+    |Lock-on| |Close      |    |check  | |distance   |
|(2.5s) | |first      |    +---+---+ +-----------+     Yes|    +---v----------+    |Friendly fire  |    |
check (radius) |    +---+-------+---+     Clear|     |Blocked         |     |    +-----v--+  +--
v--------+    |FIRE!   |  |Re-angle   |    |Fox 3 / |  |or delay   |    |EMP hot |  +-----------+
+--------+
```

15.4 Scenario Vector Pipeline

```
+-------------------+     +----------------------+     +----------------+ | MISSION TEMPLATE  |
| DYNAMIC MODIFIERS    |     | CLAMPED VECTOR | | (§10.1)          |     | (§10.2)
|     | (§10.3)        | |                  |     |                    |     |              | | [T, O, A, S, TP]  |
--> | + Event Deltas       | --> | Clamp [0,1]    | | e.g. Patrol:      |     | + Ambush:
T+0.4...   |     | Rate limit     | | [0.2,0.3,0.5,     |     | + Ally lost: T+0.2...|     | 0.4/sec
max    | |  0.2,0.1]        |     | + Intel: A-0.2...    |     | EMA smooth     | +-------------------+
+----------------------+     | (0.85 decay)   |                                                  +-------
+--------+                                                    |
+-------v--------+                                             | MANIFOLD INPUT |
| (§5.1 - §5.2) |                                             |              |
| Compute x, y   |                                               | from personality|
| + scenario     |                                              +-------+--------+
|                                          +----------+----------+
|                    |                                    +------v------+     +-------v-------+
```

```
| z = xy      |     | z = xy²       |                                        | (Linear)    |     |
(Asymmetric)  |                                        | Engagement  |     | Escalation    |
| Intensity   |     | Magnitude     |                                        +------+------+     +-------
+-------+                                        |                  |
+------v--------------------v------+                                        |      BEHAVIORAL
OUTPUT           |                                        |                              |
| - Weapon selection (§5.3)        |                                        | - Combat state
transitions (§4.2) |                                        | - Communication tone/urgency      |
| - Phrase pool selection (§7)       |                                        | - Formation
compliance           |                                        +-----------------------------------+
```

15.5 ANPC Lifecycle Diagram

```
+--------+     +------------+     +-----------------+ | SPAWN  | --> | INITIALIZE | --> |
MISSION ACTIVE  | +--------+     +------------+     +--------+--------+                |
|                | - Load identity schema   | - Scenario vector active                | - Set
personality vector | - Manifold calculations                | - Assign formation pos   | -
Phrase pool enabled                | - Calibrate manifold     | - Communication active
| - Set initial ROE        |                                +--------v--------+                              |
COMBAT STATES   |                                |                |                              |
PATROL

<- data-id="3098"> ALERT |                                |     |           |
|  ENGAGED         |                                |  /    \         |                              |
EVASIVE DAMAGED   |                                |  \    /         |                              |
RETREATING       |                                |     |           |                              |
DISABLED         |                                |     |           |                              |
```

```
DESTROYED -----> EXIT (despawn debris after 30s)                          +--------
+---------+                                  |                            (if survived)
|                          +--------v--------+                            |  MISSION END     |
|               |                            | - Reset manifold |                            |   drift
|                          | - Clear comms    |                            |   queue         |
| - Reset morale   |                            |   to base value  |
+--------+---------+                                  |                            +--------v--------+
|    DEBRIEF       |                            |               |                            | - Apply
dispo-   |                            |   sition deltas  |                            | - Update kill_
|                          |   count         |                            | - Update flight_ |
|   hours          |                            | - Trigger debrief|                            |
dialogue       |                            +--------+---------+                                  |
+------------+------------+                            |                    |                    +--------
v--------+       +--------v--------+                    |    PERSIST       |       |    DESPAWN       |
|               |       |               |                    | Named ANPCs:    |       | Procedural
ANPCs:|                    | Save all state  |       | Discard state    |                    | to
campaign     |       | Free resources   |                    | save file       |       |               |
+----------------+       +----------------+
```

Appendix A: Complete Sample ANPC — Wingman "Hotshot" (Marcus Chen)

Base Identity Fields (Document #1 §3)

Field	Value
anpc_id	"ANPC-SF-0042"
display_name	"Marcus Chen"
callsign	"Hotshot"
role	SF-WING (Wingman)
age	27
gender	Male
voice_profile	"young_male_energetic"
simulation_awareness_level	0
backstory	Marcus Chen grew up on a frontier colony where racing skimmers through canyon systems was the only entertainment. His reckless skill caught the attention of a military recruiter, and he enrolled in the fleet academy at 18. He graduated mid-class academically but top of his flight group in combat simulations. He flies with instinct over calculation and covers his fear with bravado. He lost his older sister in an early campaign engagement and channels that grief into aggressive, prove-something energy. Despite his brash exterior, he is deeply loyal to anyone who earns his trust.
disposition_toward_player	0.65 (warm, growing trust)

Personality Vector (Document #1 §4)

Axis	Value	Interpretation
Openness	0.55	Moderately open — willing to try new tactics but prefers what he knows works
Conscientiousness	0.40	Below average — impulsive, acts on instinct, sometimes ignores protocol
Extraversion	0.85	Very high — loud, talkative, feeds off attention, loves combat banter
Agreeableness	0.45	Below average — competitive, argumentative, but fiercely loyal once bonded
Neuroticism	0.35	Low-moderate — generally stable, but grief can surface under extreme stress

Starfighter Extended Fields (§3.1)

Field	Value
ship_class	Interceptor
squadron_id	"SQ-07"
squadron_name	"Shadow Wolves"
combat_rating	0.72
weapon_loadout	["WPN-LAS", "WPN-SCT", "WPN-PTN"]
flight_hours	1,847
kill_count	38
loyalty_score	N/A (allied military — not allegiance-

Field	Value
	fluid)
damage_threshold	0.35 (stays in longer than most — overconfidence)
preferred_tactic	aggressive
formation_position	wing_left

Manifold Parameters

Parameter	Value	Derivation
Base aggression impulse (x_0)	0.55	1.0 - Agreeableness = 0.55
Volatility baseline (y_0 for $z=xy^2$)	0.35	Neuroticism = 0.35
Adrenaline spike magnitude	+0.25	High Extraversion amplifies initial combat rush
Fatigue resistance	0.7	Low Conscientiousness but high energy — moderately fatigue-resistant
Morale floor	0.20	Grief-driven determination — will not completely break

Phrase Pool Assignment

Character-Pool: "Hotshot" (§7.2) — overrides 5 Title-Pool phrases with brash, confident variants. All non-overridden phrases fall through to Title-Pool (§7.1), then Universal-Pool (Document #1 §8).

Appendix B: Complete Sample ANPC — Enemy Ace "Nightshade" (Unknown)

Base Identity Fields (Document #1 §3)

Field	Value
anpc_id	"ANPC-SF-E-0001"
display_name	"Unknown" (revealed as "Kael Draven" if player reaches Appendix B narrative trigger)
callsign	"Nightshade"
role	SF-EACE (Enemy Ace)
age	Unknown (estimated 35–40)
gender	Male
voice_profile	"mature_male_calm_menacing"
simulation_awareness_level	0
backstory	Nightshade is a ghost in enemy intelligence files — no confirmed identity, no known origin. He appeared in combat theaters three years ago piloting a matte-black Heavy Fighter with nonstandard modifications. His kill count is estimated at 100+ confirmed. He operates outside the standard enemy chain of command, appearing at critical engagements and vanishing after. Rumors suggest he was once an allied pilot who defected after a classified incident. He fights with surgical precision and cold professionalism, breaking his silence only to address pilots who impress him — or to deliver final words to those who don't.
disposition_toward_player	-0.60 (hostile but curious — he respects worthy opponents)

Personality Vector (Document #1 §4)

Axis	Value	Interpretation
Openness	0.65	Moderately high — adapts tactics mid-combat, unpredictable approach angles
Conscientiousness	0.85	Very high — methodical, patient, waits for the perfect shot
Extraversion	0.30	Low — silent for long stretches, speaks only with purpose
Agreeableness	0.15	Very low — merciless in combat, no quarter asked or given
Neuroticism	0.20	Very low — ice-cold under pressure, almost unnervingly calm

Starfighter Extended Fields (§3.1)

Field	Value
ship_class	Heavy Fighter (custom — "Nightshade Special")
squadron_id	"SQ-X" (unaffiliated — operates solo)
squadron_name	None
combat_rating	0.94
weapon_loadout	["WPN-LAS", "WPN-SCT", "WPN-PTN", "WPN-EMP"]
flight_hours	12,000+ (estimated)
kill_count	100+ (confirmed estimate)
loyalty_score	N/A (enemy — loyalty is to himself)

Field	Value
damage_threshold	0.15 (extremely resilient — fights until nearly destroyed)
preferred_tactic	ambush
formation_position	free_roam (operates solo)

Surrender Conditions

Nightshade will surrender only when ALL of the following conditions are met simultaneously:

- Morale < 0.15 (morale floor — his is 0.10 due to extreme low Neuroticism)
- Hull integrity < 0.08
- Player has personally dealt > 50% of total damage to Nightshade
- No escape vector available (surrounded or engines disabled)

If surrender triggers, Nightshade broadcasts on CH-ENM (intercepted): "...You're the first. Well fought." If the player then holds fire, a narrative event unlocks revealing his identity.

Unique Combat Behaviors

- **Signature maneuver: "Nightshade Fade"** — drops to zero emissions (disappears from sensors for 4 seconds), repositions behind the target, then attacks from a blind spot. Triggered when hull < 0.6 and player is pursuing.
- **Target priority:** Ignores all targets except the player unless directly threatened by another ANPC. +0.5 bias to player at all times.
- **Taunt escalation:** Taunts become more respectful as the player proves their skill. At hull < 0.3, taunts shift to grudging admiration.

Appendix C: Complete Sample ANPC — Command Base Operator "Lighthouse" (Dr. Amara Okafor)

Base Identity Fields (Document #1 §3)

Field	Value
anpc_id	"ANPC-SF-0003"
display_name	"Dr. Amara Okafor"
callsign	"Lighthouse"
role	SF-CMDOP (Command Base Operator)
age	42
gender	Female
voice_profile	"mature_female_calm_authoritative"
simulation_awareness_level	0
backstory	Dr. Amara Okafor was a stellar cartographer before the war pulled her into military intelligence. Her ability to read sensor data and predict enemy movements made her the fleet's most valuable operations officer. She has never fired a weapon in combat but has guided more pilots home than any other operator in the fleet. She takes every pilot loss personally and maintains a private memorial list. Her callsign "Lighthouse" was given by pilots who say her voice in their headset is like seeing a lighthouse through a storm — steady, guiding, and a promise that the shore is close. She is deeply analytical but carries warmth beneath her professional exterior.
disposition_toward_player	0.70 (supportive, invested in player's survival and growth)

Personality Vector (Document #1 §4)

Axis	Value	Interpretation
Openness	0.75	High — creative in problem-solving, sees unconventional solutions in sensor data
Conscientiousness	0.90	Very high — meticulous with intel, never provides unverified information
Extraversion	0.40	Moderate-low — speaks when needed, economical with words during combat
Agreeableness	0.70	High — supportive, nurturing, but will push back on reckless plans
Neuroticism	0.10	Very low — unshakable under pressure; her calm is the anchor for the squadron

Starfighter Extended Fields (§3.1)

Field	Value
ship_class	N/A (operates from command base — non-combatant)
squadron_id	"CMD-01"
squadron_name	"Central Command"
combat_rating	N/A (non-combatant)
weapon_loadout	N/A
flight_hours	0 (ground-based)
kill_count	0
loyalty_score	N/A (allied military)
damage_threshold	N/A

Field	Value
preferred_tactic	support
formation_position	N/A

Communication Profile

Attribute	Value
Primary channel	CH-FLT (fleet-wide), CH-SQD (direct squadron support)
Communication frequency	Event-driven — mission briefings, intel updates, threat warnings, extraction coordination
Communication style	Calm, precise, data-rich. Uses coordinates, ETAs, and threat-level assessments. Never raises voice except during imminent-danger warnings (collision, ambush).
Private channel usage	Uses CH-PVT to check on player after high-stress missions. Unlocks personal dialogue at disposition > 0.75.
Signature pattern	Always provides three pieces of information: What, Where, When. Example: "Enemy flight, bearing 045, contact in 90 seconds."

Appendix D: Mission Script Integration Example

The following walkthrough demonstrates a complete patrol-to-ambush mission integrating all ANPC systems defined in this document.

Phase 1 — Mission Start: ANPCs Spawn with Patrol Scenario Vector

Event	System Action
Mission loads: "Patrol Sector 7"	Scenario vector set to Patrol template: [0.2, 0.3, 0.5, 0.2, 0.1]
Player spawns at carrier	Player ship initialized at launch position
Wingmen spawn: Hotshot (wing_left), Frostbite (wing_right)	Identity schemas loaded; manifold calibrated against patrol scenario vector. Hotshot's initial z=xy: 0.55 × 0.3 = 0.165 (very low intensity). Formation: V-Formation.
Lighthouse opens comms	SF-MC-001 triggered: "All ships, this is Lighthouse. Patrol Sector 7 — standard sweep. Report any contacts. Lighthouse out."
Hotshot acknowledges	HS-MC-001 (custom): "Hotshot's ready to roll. Let's see if anyone's out there."
Frostbite acknowledges	FB-MC-001 (custom): "Copy."
Combat state: CS-PATROL for all ANPCs	Communication frequency: casual (12–20s intervals). Morale: 0.7 (baseline).

Phase 2 — Contact Detected: Vector Shifts, ANPCs Transition States

Event	System Action
T+120s: Sensor detects 4 unidentified contacts at 2000m	Ambiguity component surges. Dynamic modifier: "New contacts" → Ambiguity +0.2. Scenario vector: [0.2, 0.3, 0.7, 0.2, 0.1]
Hotshot reports	SF-CE-005 triggered: "Hotshot, bogey at eleven o'clock low, 2000 meters, over."
All ANPCs transition: CS-PATROL → CS-ALERT	Shields raised, weapons armed. Formation tightens (formation_tightness auto-adjusts from

Event	System Action
	0.5 to 0.7).
T+125s: Contacts classified as Bandit (4 enemy Interceptors)	Dynamic modifier: "Ambush detected" → Threat +0.4, Ambiguity -0.3, Time_Pressure +0.3. Vector: [0.6, 0.3, 0.4, 0.3, 0.4]. EMA smoothing begins.
Frostbite reports	FB custom: "Four bandits. Interceptors."
Lighthouse relays intel	"Lighthouse confirms — four hostile interceptors, combat rating estimated medium. No additional contacts on long-range scan."

Phase 3 — Engagement: Manifold Drives Weapon Selection and Aggression

Event	System Action
Player sets ROE to Weapons Free	ROE-FREE propagates to Hotshot and Frostbite. Both acknowledge.
ANPCs transition: CS-ALERT → CS-ENGAGED	Adrenaline spike: Hotshot's volatility +0.25 (decays over 30s). Frostbite's volatility +0.10 (low Neuroticism dampens spike).
Hotshot's manifold lookup	x = 0.55 + 0.25 (order boost) = 0.80; y = Opportunity 0.5 (enemy exposed). z=xy = 0.40. Weapon: Lasers sustained fire. Hotshot engages nearest bandit.
Frostbite's manifold lookup	x = 0.30 (low aggression); y = 0.5. z=xy = 0.15. Weapon: Lasers probing. Frostbite takes defensive position, engaging at range.
Hotshot fires	SF-CE-003 triggered: "Guns, guns, guns!" Character-pool override: "Watch this!"
Frostbite fires	FB custom: "Engaging." (minimal, per character pool)
T+135s: Hotshot splashes enemy #1	SF-CE-004 triggered. Character-pool: "Boom! That's how it's done! Add another to the board!" Morale +0.10 (self) + 0.05 × 0.3 contagion to

Event	System Action
	Frostbite.

Phase 4 — Ally Damaged: Morale Effects, Disposition Shifts, Phrase Triggers

Event	System Action
T+150s: Frostbite takes concentrated fire from 2 bandits	Frostbite transitions: CS-ENGAGED → CS-EVASIVE. Hull drops to 0.40 (below damage_threshold of 0.45).
Frostbite reports	FB-DR-001: "Shields gone." (character pool: minimal)
Morale impact	Frostbite morale: -0.05/sec (sustained fire). Hotshot morale: -0.05 × 0.3 contagion = -0.015.
Player engages Frostbite's attacker, destroys it	Disposition shift: Frostbite → player +0.15 ("player saves ANPC from destruction"). Frostbite was in CS-EVASIVE with hull < 0.5 — trigger met.
Frostbite acknowledges rescue	FB custom private channel: "...Noted." (Frostbite's version of gratitude — disposition increases but dialogue remains minimal)
Remaining bandits: morale check	Enemy morale: 2 of 4 destroyed → morale modifier -0.15 × 2 = -0.30. Remaining enemies at morale 0.40 — medium range, continue fighting but defensive.

Phase 5 — Victory: Post-Combat Recovery, Debrief Dialogue

Event	System Action
T+180s: Final enemy destroyed (by Hotshot)	All hostiles eliminated. Hotshot: kill_count +2 this mission. Hotshot

Event	System Action
	morale restored to 0.75 (kills + victory).
ANPCs transition: CS-ENGAGED → CS-PATROL	30-second manifold drift reset begins. Scenario vector decays toward patrol template via EMA smoothing.
Frostbite reformation	Frostbite transitions CS-EVASIVE → CS-PATROL (via CS-DAMAGED intermediate, hull self-stabilizes). Reforms to wing_right position.
Squadron Leader (NPC) orders reform	SF-TC-005: "Hostiles clear. Reform on me, V-Formation."
Hotshot post-combat	SF-MB-005 override: "Ha! Two for me. Anyone else want to argue about who buys drinks tonight?" (morale high, Extraversion high)
Frostbite post-combat	FB custom: "Hull at forty percent. Functional." (no celebration, damage-focused)
Lighthouse mission complete	"Lighthouse to patrol group — all contacts neutralized. Sector 7 clear. Return to base. Good work. Lighthouse out."
Mission end — persistence writes	Hotshot: kill_count += 2, flight_hours += 0.5, disposition_toward_player += 0.04 (cooperative kills). Frostbite: disposition_toward_player += 0.15 (rescued) + 0.03 (survived together) = +0.18 total. Manifold drift resets. Morale resets to base values.

Appendix E: Difficulty Scaling Matrix

The following table defines how all tunable ANPC parameters change across difficulty levels. These values are applied as multipliers or overrides to the base parameters defined in §11.3.

Allied ANPC Scaling (Player's Wingmen)

Parameter	Easy	Normal	Hard	Veteran
accuracy_curve	0.70	0.60	0.50	0.40
reaction_time (seconds)	0.3	0.5	0.7	1.0
evasion_skill	0.70	0.50	0.35	0.25
morale_sensitivity	0.7	1.0	1.3	1.6
damage_threshold (override)	0.25	0.35	0.45	0.55
aggression_multiplier	1.2	1.0	0.8	0.7
comms_frequency_multiplier	1.3	1.0	0.8	0.6

Enemy ANPC Scaling

Parameter	Easy	Normal	Hard	Veteran
accuracy_curve	0.30	0.50	0.70	0.85
reaction_time (seconds)	1.2	0.7	0.4	0.2
evasion_skill	0.20	0.40	0.60	0.80
morale_sensitivity	1.5	1.0	0.7	0.5
combat_rating (base range)	0.2 – 0.4	0.3 – 0.6	0.5 – 0.8	0.7 – 0.95
aggression_multiplier	0.7	1.0	1.3	1.6
formation_tightness	0.3	0.5	0.7	0.9
target_switch_	5.0	3.0	2.0	1.0

Parameter	Easy	Normal	Hard	Veteran
cooldown (seconds)				

Enemy Ace Scaling (SF-EACE)

Parameter	Easy	Normal	Hard	Veteran
combat_rating	0.70	0.82	0.90	0.96
accuracy_curve	0.55	0.70	0.82	0.92
reaction_time	0.5	0.3	0.2	0.1
damage_threshold	0.25	0.18	0.12	0.08
morale_floor	0.25	0.15	0.10	0.05
signature_maneuver_frequency	Every 45s	Every 30s	Every 20s	Every 12s

Global Mission Scaling

Parameter	Easy	Normal	Hard	Veteran
Max concurrent enemies per encounter	3	5	7	10
Enemy reinforcement probability	10%	25%	45%	70%
Allied reinforcement probability	50%	30%	15%	5%
Enemy ace appearance rate	1 per 4 missions	1 per 3 missions	1 per 2 missions	Every mission
Player shield	1.5	1.0	0.8	0.6

Parameter	Easy	Normal	Hard	Veteran
recharge rate multiplier				

DESIGN NOTE

The difficulty scaling philosophy follows the ButterflyFX™ principle: difficulty should feel like flying against smarter, more coordinated opponents — not like absorbing more damage from identical ones. On Veteran, enemies fly in tighter formations, switch targets faster, use maneuvers more frequently, and have much lower morale sensitivity (they do not break easily). Allied ANPCs become slightly less effective, creating a genuine sense of earned victory. The manifold surfaces handle this naturally: higher enemy combat_rating and aggression_multiplier shift their manifold lookup points toward higher-z regions, producing more intense, committed combat behavior without any special-case code.

Starfighter ANPC Guideline Document — Document #2

ButterflyFX™ ANPC Design Suite | Version 1.0 | April 2026

Author: Kenneth Bingham | "Store Geometry Not Bytes"

Manifold Surfaces $z=xy$ and $z=xy^2$ — Foundational Data-as-Geometry System

This document inherits from and must not contradict Document #1 (General ANPC Guideline Document).

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